Interpreting FCFS Appointment Simulation Results Under the Current Formulation

Metric behavior, access losses, and model checks

CUIMC Appointment Simulation

May 2026

Abstract

This note interprets simulation outputs from a discrete-event FCFS appointment model. The goal is to understand what the outputs show and whether they are mechanically plausible under the current formulation—not to make empirical claims about real clinics.

The central finding is that utilization and served rate separate sharply. At baseline ($\lambda = 100$ arrivals per day, 32 slots per day, 14-day horizon), average utilization is 0.839 while overall served rate is 0.269. This separation is mechanically plausible because the two metrics have different denominators: utilization is normalized by slots, served rate by arrivals. The baseline is structurally oversubscribed. Patients are indeed served on future days (the horizon extends to $H = 14$ days), but each service day has exactly 32 slots regardless of which arrival cohort fills them. Over any measurement window of D days, at most $32D$ visits can be completed against approximately $100D$ arrivals, so served rate cannot exceed $32/100 = 0.32$.

Arrival outcome decomposition at baseline shows that balking (35.6% of arrivals) and cancellation (32.3%) are the dominant loss channels, not horizon saturation: no-offer is 0 at baseline demand. The high cancellation fraction is mechanically consistent with the long average booking delay (8.35 days) combined with a 10% daily cancellation probability.

Mean offered wait is preferable to mean accepted wait as a delay measure, because offered delay is recorded before the balking decision and therefore includes patients who ultimately reject the offer. Offered wait still excludes no-offer patients, so it must be read together with the no-offer share—which is zero at baseline but grows above $\lambda \approx 110$.

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1 Simulation Mechanics Relevant for Interpretation

The following mechanics directly affect how outputs should be read. The full formulation is in the simulation documentation.

Finite booking horizon and FCFS pooling. Each day, arrivals for both classes are pooled and randomly permuted into one FCFS sequence. Each patient is offered the earliest available slot within the H-day horizon (including same-day, $r = 0$). Because FCFS operates on a permuted joint sequence, class labels are irrelevant to slot assignment: neither class is prioritized. Class disparities in outcomes arise only from differences in arrival rates or behavioral parameters.

Offered delay τ recorded before balking. When a valid slot exists, the offered delay τ is recorded before the patient decides whether to accept. Mean offered wait therefore includes patients who ultimately balk. Mean accepted wait excludes them, which biases it downward when high-τ patients disproportionately balk.

No-show based on original τ. No-show probability is a function of the original offered delay τ, not of the remaining residual day r at service time. A patient who accepted a slot 9 days out carries their high-delay no-show probability regardless of when they end up being seen.

Future cancellations only; no rebooking of no-show slots. Cancellations apply only to future appointments ($r \geq 1$). Same-day cancellations are not modeled. Slots vacated by no-shows are not rebooked—once a patient no-shows, that slot is lost.

No-offer rule. If the horizon is full when a patient reaches the front of the FCFS queue, that patient and all remaining arrivals for that day are counted as `no_offer`. This means a single saturation event affects the entire remainder of the day's queue, which could introduce daily-order artifacts.

Utilization denominator. Utilization counts completed visits per available slot, not scheduled appointments per slot. No-shows, cancellations, and unbooked slots all reduce utilization. A high-utilization day is one where most slots became completed visits.

2 Baseline Setup and Diagnosis

2.1 Parameter Setup

The baseline (`configs/baseline.yaml`) parameters are listed in Figure 1. The key feature is the demand-to-capacity ratio: 100 arrivals per day against 32 slots per day is a 3:1 oversubscription, well above any realistic clinic load. The purpose is to probe how the metrics behave across a wide range, not to model a plausible clinic.

Parameter	Value
Class 1 arrival rate λ_1 (per day)	50
Class 2 arrival rate λ_2 (per day)	50
Total arrival rate λ (per day)	100
Slots per day S	32
Booking horizon H (days)	14
Cancellation probability ϕ	0.10
Balking threshold θ^{balk} (days)	9
High-delay balking prob. b^{high}	0.50
Low-delay balking prob. b^{low}	0.00
No-show threshold $\theta^{\text{no-show}}$ (days)	6
High-delay no-show prob. ξ^{high}	0.30
Low-delay no-show prob. ξ^{low}	0.00

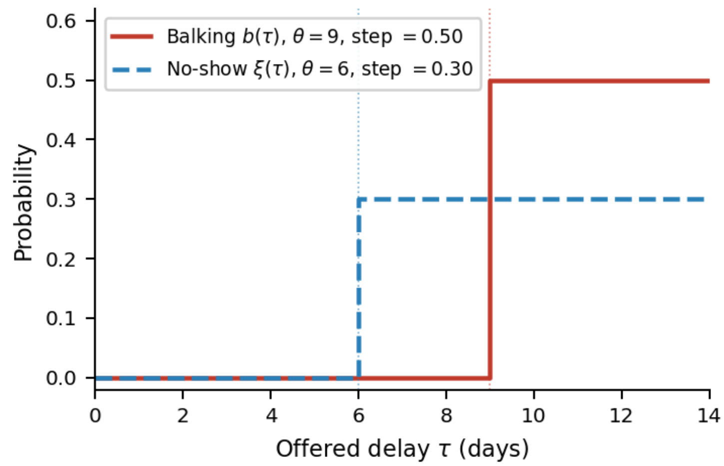

Figure 1: Left: baseline parameter values (`configs/baseline.yaml`), symmetric across both classes; measurement window 365 days after a 30-day burn-in. Right: balking and no-show step functions at baseline. Both probabilities are 0 below their respective thresholds and jump to 0.50 (balking) and 0.30 (no-show) above them.

2.2 Aggregate Metrics at Baseline

Metrics are averaged over a single run of 365 measurement days (seed 11), preceded by a 30-day burn-in to reach steady state and followed by a 10-day cooldown to allow late bookings to resolve. All figures in Table 1 are point estimates from that single run; the sensitivity appendix uses multiple seeds (2027–2029 for slices, 4101–4102 for the regression screen) to assess stability and the influence of each parameter.

Metric	Value
Average utilization	0.839
Overall served rate	0.269
Mean offered wait (days)	9.30
Mean accepted wait (days)	8.35
Overall balking rate	0.356
Class served-rate gap (Δ_{access})	0.001

Table 1: Baseline aggregate metrics. The near-zero class gap is expected: both classes have identical behavioral parameters and symmetric arrival rates, so class labels are exchangeable under pooled FCFS.

2.3 The Utilization–Access Gap

$$\text{average utilization} = \frac{\text{completed visits}}{\text{available slots}}, \qquad \text{overall served rate} = \frac{\text{completed visits}}{\text{arrivals}}.$$

Utilization (0.839) and served rate (0.269) differ by a factor of roughly 3.1. This is mechanically expected because the denominators differ. Utilization is normalized by supply; served rate is normalized by demand. When demand far exceeds supply, served rate is bounded by the supply-to-demand ratio; utilization is not.

At baseline, $100/32 = 3.125$ arrivals per slot. The ceiling argument is a flow constraint: patients are served on future days (via the booking horizon), but each service day has exactly 32 slots regardless of which arrival cohort fills them. Over D days at most $32D$ visits can complete against $100D$ arrivals, so served rate cannot exceed $1/3.125 = 0.32$. The observed served rate of 0.269 falls below this ceiling because no-shows consume slots without producing completed visits.

The simulation implies, under these assumptions, that a facility-level utilization measure would give a misleading picture of patient access at this demand-to-capacity ratio.

3 Arrival Outcome Decomposition

3.1 Loss Channels at Baseline

Every arrival ends up in exactly one outcome state. At baseline:

Outcome	Share of arrivals	Interpretation
Served	0.269	Completed visit
Balked	0.356	Received an offer, rejected the delay
Canceled	0.323	Booked, but canceled before service day
No-show	0.052	Kept booking, did not attend
No-offer	0.000	Horizon full; no slot offered
Unresolved	<0.001	Booked but not resolved before simulation end
Total	1.000	

Table 2: Arrival outcome decomposition at baseline ($\lambda = 100$). No-offer is zero: the horizon does not completely fill at baseline demand because cancellations continuously free future capacity.

Two findings stand out. First, the no-offer rate is zero even at 100 arrivals/day. The horizon does not saturate because cancellations (10%/day for each future appointment) continuously free slots. However, cancellation frees far-future capacity, which is then offered to new arrivals at long delays—reinforcing the same long-delay dynamic that drives balking.

Second, the 32.3% cancellation rate (as a share of arrivals) is not self-evidently wrong. Of the 64.4% of arrivals that book an appointment, roughly half ultimately cancel. With an average booking delay of 8.35 days and a 10% daily cancellation probability, the implied cumulative cancellation probability is $1 - 0.9^{8.35} \approx 0.59$. The observed cancellation rate among booked patients is in the same range, though somewhat lower—possibly because some booked patients face fewer cancellation-check days when scheduled soon.

Inspection of `simulation/engine.py` confirms that the engine applies a fresh Bernoulli draw to every future appointment on each simulated day, so the cumulative argument is consistent with the implementation.

3.2 How Loss Channels Shift with Demand

Figure 2 shows the arrival outcome decomposition as total daily arrivals vary from 30 to 170. At low demand ($\lambda \lesssim 50$), virtually all arrivals are served. As demand rises above the 32-slot daily capacity, the served rate falls. The no-offer rate remains zero up to $\lambda \approx 110$, at which point the horizon begins to saturate.

The transition from a cancellation- and balking-dominated regime (below $\lambda \approx 110$) to a horizon-saturation regime (above) is a structural feature of the model. It suggests that reporting the no-offer share alongside served rate and offered wait is necessary to understand which failure mode dominates in a given parameter setting.

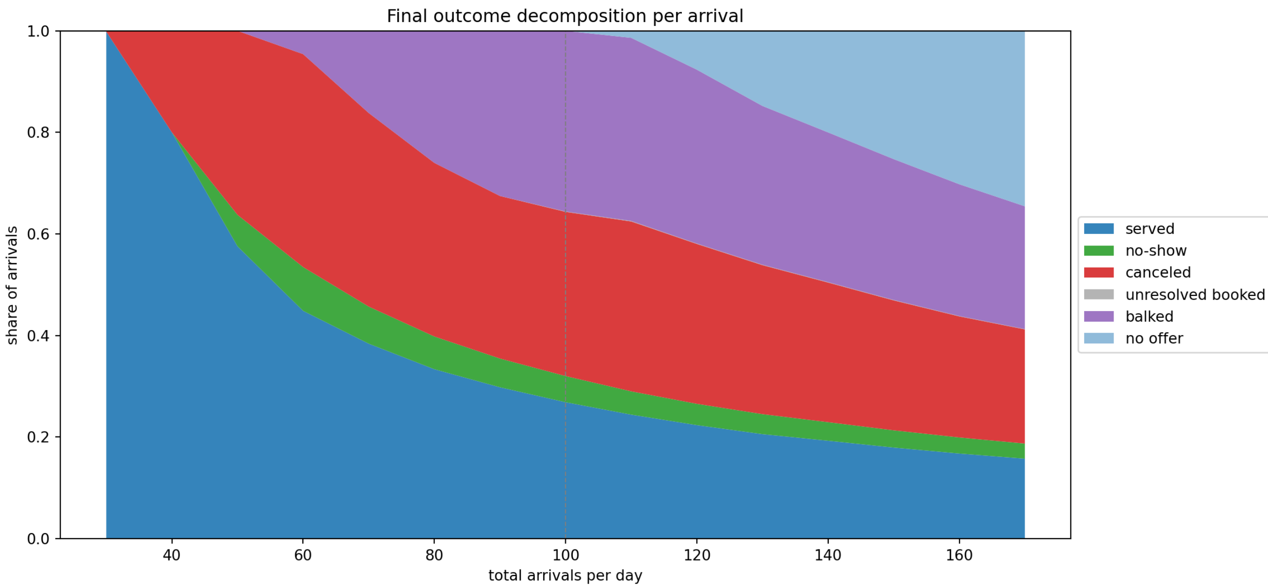

Figure 2: Arrival outcome decomposition as total daily arrivals vary from 30 to 170. At baseline ($\lambda = 100$, marked by a dashed vertical line), balking and cancellation are the dominant loss channels; no-offer is zero. No-offer becomes substantial only above $\lambda \approx 110$.

4 Cross-Metric Diagnostics

Figure 3 shows three diagnostics that together capture the core structure of the simulation output.

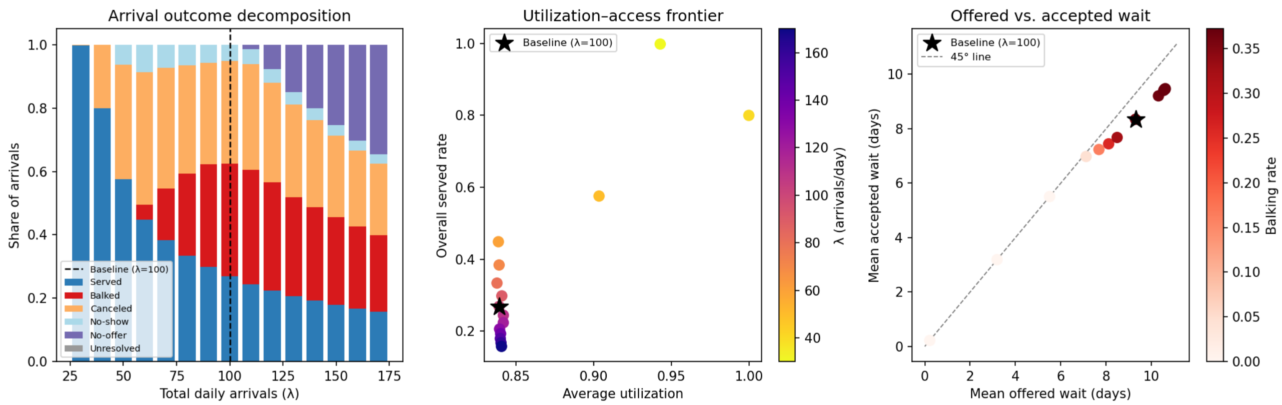

Figure 3: *Left:* Arrival outcome decomposition as total daily arrivals vary. At baseline ($\lambda = 100$, dashed line), balking and cancellation are the dominant loss channels; no-offer is zero. No-offer appears only above $\lambda \approx 110$. *Center:* Utilization–access frontier. The plot reveals two regimes separated by the onset of balking. When balking is active ($\lambda \geq 60$, the main cluster), utilization is pinned near 0.84 while served rate falls from 0.45 to 0.16. When balking is absent ($\lambda \leq 50$, the three isolated points), both metrics co-move. *Right:* Offered vs. accepted wait. Points below the 45° line indicate that accepted wait understates the delay patients face. The gap widens with balking rate (red intensity).

1. Utilization–access frontier (center panel). The center panel shows two distinct regimes, cleanly separated by whether balking is active. In the balking regime ($\lambda \geq 60$), utilization is pinned near 0.84 while served rate collapses from 0.45 to 0.16: as one patient balks or cancels, a waiting patient fills the slot, sustaining the slot-fill rate even as fewer arrivals ultimately receive service. In the no-balking regime ($\lambda \leq 50$), that queue does not exist, and the two metrics co-move. The apparent decoupling in the cluster is therefore not a general property of the metrics; it is a consequence of balking providing a surplus of waiters to absorb every freed slot.

2. Offered vs. accepted wait (right panel). All points fall below the 45° line: accepted wait consistently understates offered wait. The gap widens as balking rate increases (red intensity). This is mechanically consistent: patients who balk tend to have above-average τ, so removing them from the accepted sample pulls the mean down. Any policy or parameter change that raises balking will make accepted wait appear to improve while offered wait and served rate both worsen.

3. No-offer and the boundary of offered-wait validity. At baseline, no-offer is zero, so offered wait covers all patients who received any delay information. Above $\lambda \approx 110$, a growing share of arrivals receive no offer and are excluded from offered wait entirely. At $\lambda = 170$, the no-offer share reaches approximately 34.6% of arrivals. In that regime, offered wait alone misses the worst-off patients.

5 Metric Hierarchy Under This Formulation

Metric	What it measures	Mechanical interpretation	Main caveat
Overall served rate	Arrivals → completed visits	Primary access measure; bounded above by S/λ	Does not decompose loss channels
Average utilization	Completed visits per available slot	Capacity-use measure; decoupled from served rate only when balking is active and a surplus of waiters refills every freed slot	High value does not imply good access
Mean offered wait	Delay among patients who received any offer	Less selective than accepted wait; includes patients who then balk	Excludes no-offer patients entirely
Mean accepted wait	Delay among patients who accepted bookings	Conditional; pulled down when high-delay patients balk	Can fall as access worsens
Balking rate	Share of offered patients who reject	Congestion diagnostic	Not a success metric
Class served-rate gap	Class 1 served rate minus Class 2	Meaningful only when classes differ in behavior or arrivals	Near-zero in symmetric baseline

Table 3: Metric interpretation under the current FCFS formulation.

5.1 Balking-Mediated Decoupling of Utilization and Access

The center panel of Figure 3 separates into two regimes with a clean boundary at the onset of balking. Every point with $\lambda \geq 60$ has a positive balking rate (ranging from 4.6% to 36.1%) and utilization constrained to 0.839–0.841; every point with $\lambda \leq 50$ has zero balking, and both utilization and served rate vary freely.

In the balking regime, utilization is pinned near 0.84 while served rate falls monotonically from 0.45 to 0.16 as demand rises from 60 to 170. This is a direct consequence of how cancelled slots are filled: as long as a surplus of balking-tolerant patients is waiting, each freed slot is immediately claimed, keeping the slot-fill rate near a no-show-adjusted ceiling regardless of how many arrivals ultimately fail to receive service. The apparent decoupling of utilization from served rate holds only within this regime—it is a property of the balking mechanism, not of the metrics themselves.

The three no-balking points ($\lambda \in \{30, 40, 50\}$) tell a different story. At $\lambda = 30$, demand is below daily capacity; virtually all arrivals are served (served rate = 0.998) and utilization reaches 0.943, though the horizon does not fully saturate from a single day's arrivals. At $\lambda = 40$, slots are completely absorbed (utilization = 1.000), but 20% of arrivals cancel before their service day; with no queue to refill those slots, served rate falls to 0.800. At $\lambda = 50$, demand exceeds daily capacity yet no patient balks; a growing cancellation chain that goes unrefilled pulls utilization back to 0.903 and served rate to 0.576. Across all three, utilization and served rate co-move rather than decouple, confirming that balking—not a structural property of the model—is what sustains high slot fill under declining access.

Figure 4 extends the comparison to $\lambda \leq 110$ and plots both mean offered and mean accepted delay against the two behavioral thresholds. In the no-balking regime ($\lambda \leq 50$) the two series coincide: every offer is accepted, so accepted and offered delay are the same by construction. Mean accepted delay rises from 0.22 days at $\lambda = 30$ to 5.50 days at $\lambda = 50$—in all three cases strictly below the balking threshold ($\theta^{\text{balk}} = 9$ days), confirming that zero balking is self-consistent. The $\lambda = 50$ point sits just below the no-show threshold ($\theta^{\text{no-show}} = 6$ days), however, which is why cancellation losses are elevated there: although the mean does not breach 6 days, a non-trivial share of the delay distribution does.

The two series split at exactly $\lambda = 60$, the onset of balking. From that point onward offered delay continues to rise steeply while accepted delay grows more slowly, because patients offered the longest waits are precisely those who balk, selectively trimming the high-τ tail from the accepted sample. By $\lambda = 110$ the gap reaches roughly 1.1 days; offered delay crosses the balking threshold while accepted delay has not yet done so, illustrating why accepted wait understates congestion in high-balking regimes.

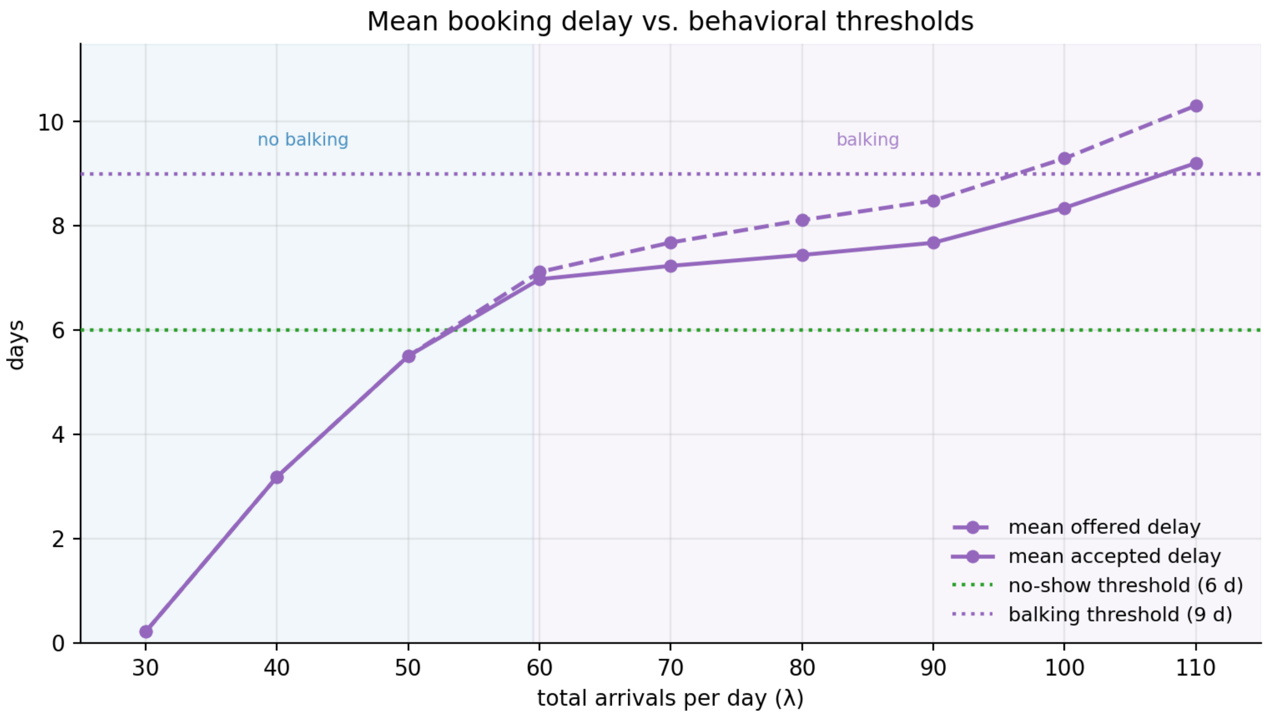

Figure 4: Mean offered (dashed) and accepted (solid) booking delay from $\lambda = 30$ to $\lambda = 110$, plotted against the no-show threshold (6 days, green) and balking threshold (9 days, purple). Shading marks the no-balking ($\lambda \leq 50$, blue) and balking ($\lambda \geq 60$, purple) regimes. The two series coincide where no one balks and diverge thereafter, with the gap widening as balking removes the high-τ tail from the accepted sample.

5.2 Offered Wait vs. Accepted Wait

The right panel of Figure 3 shows that accepted wait is consistently below offered wait across all demand levels, with the gap widening as balking rate rises; Figure 4 shows the same split in absolute terms as a function of λ.

The implication is that accepted wait understates congestion in high-balking regimes. Offered wait is the better denominator for any congestion diagnostic—but it still excludes no-offer patients, who receive no delay information at all.

5.3 No-Offer as a Distinct Access Failure

Mean offered wait is computed among patients who received a slot offer. When the horizon saturates ($\lambda \gtrsim 110$ in this formulation), a growing share of arrivals receive no offer and are entirely absent from the offered-wait calculation. Figure 4 covers $\lambda \leq 110$, the range immediately preceding this saturation regime; at $\lambda = 110$ offered delay has just crossed the balking threshold and no-offer patients are beginning to accumulate. A useful diagnostic is to report no-offer share alongside offered wait. Together they describe both the level and the completeness of delay information.

6 Sensitivity Screen: Main Patterns

The one-at-a-time sensitivity runs and the regression screen are described in Appendix B and D. The patterns most relevant for interpretation are summarized here.

Demand pressure. Total arrival rate has the largest absolute standardized regression coefficient for overall served rate ($\beta = -0.774$) and is the dominant driver of offered wait ($\beta = +0.576$). This is mechanically consistent: more arrivals compete for a fixed number of slots, pushing offers farther out and reducing access.

No-show and cancellation. Average no-show threshold has the largest coefficient for utilization ($\beta = +0.515$): a higher threshold delays the onset of high no-show probability, keeping more booked slots as completed visits. Cancellation has a positive coefficient for served rate ($\beta = +0.116$) and a negative one for offered wait ($\beta = -0.442$). The positive sign on served rate is counterintuitive at first: the mechanism is that cancellations free future capacity, which can absorb more patients and slightly improve the served fraction—a rebooking-chain effect under FCFS.

The mechanism is confirmed by the engine order and the arrival outcome decomposition (Figure 5). Cancellations fire at the start of each day (step 1 of the daily loop), freeing future slots before new arrivals are generated. New arrivals then book in FCFS order (step 3–5 of the same day), so every freed slot is immediately available for rebooking—no slot goes empty as a result of cancellation alone. The net effect on overall served rate is positive because: (i) the replacement arrival often books a nearer slot, reducing its no-show exposure; and (ii) at baseline load there is always a surplus of unserved arrivals to fill freed capacity.

The class-level decomposition in Figure 5 makes the asymmetry explicit. As one class's cancellation probability rises, that class's own served rate falls sharply—its patients cancel themselves out. The other class's served rate rises by a comparable amount, capturing the freed slots. The net effect on overall served rate is small but consistently positive across the full cancellation range. The effect would not hold at very low demand, where freed slots might go unfilled; the positive coefficient reflects the high-load regime that dominates the regression sample.

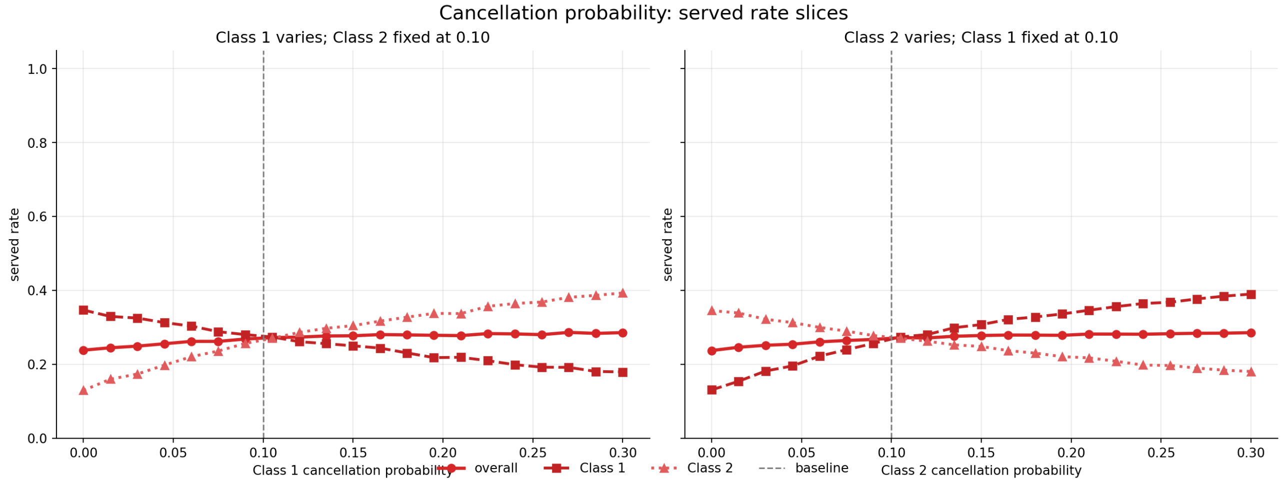

Figure 5: Served rate as a function of cancellation probability (one class varies, the other held at baseline $\phi = 0.10$). The canceling class loses its own served rate; the other class gains by filling freed slots. Overall served rate (circles) rises modestly, confirming the rebooking-chain mechanism. Dashed vertical line marks the baseline.

Balking and no-show step. The reparametrization into level and difference reveals two distinct effects. The combined step level—the average of the balking and no-show high-delay probabilities—carries a negative coefficient for utilization ($\beta = -0.288$), served rate ($\beta = -0.106$), and offered wait ($\beta = -0.181$). Higher behavioral intensity in either direction reduces slot fill and access; the wait reduction is a selection effect (high-step patients exit the queue), not a congestion improvement.

Balk vs. no-show dominance. The difference (balking step minus no-show step) is significant for utilization ($\beta = +0.400$) and served rate ($\beta = +0.186$). When balking exceeds no-show, patients who cannot obtain near-term slots preemptively reject rather than accepting a distant booking and then missing it. The net effect is that fewer booked slots go to waste: the replacement arrival who takes a freed near-term slot tends to have lower no-show exposure, while a no-show-dominant regime wastes already-booked capacity with no equivalent rebooking benefit. The sign alone is not significant in any target; the effect is driven by the magnitude of the dominance, not a threshold.

Mediation analysis confirms that this path operates almost entirely through the no-show rate: the coefficient on `balk_minus_noshow_step` drops from 0.284 to 0.088 once no-show rate is added as a covariate, with a residual direct effect remaining. Figure 6 shows this structure directly. The heatmap gradient for utilization runs almost entirely along the no-show step axis: rows at the top (high no-show step) are uniformly dark regardless of balking step, while the steepest gains from balking step appear only in the lower half of the heatmap where no-show step is modest. The top projection (fix no-show step = 0.3, vary balking step) confirms the near-flat response of utilization to balking step alone. The bottom projection (fix balking step = 0.5, vary no-show step) shows the steep monotone decline that no-show step imposes. The diagonal (`balk_step` = `noshow_step`) marks the boundary between the two dominance regimes; the baseline parameter point (balking step = 0.50, no-show step = 0.30) sits below it, in balk-dominant territory. The deep-dive analysis (Section 7) confirms that accepted wait can fall while served rate also falls, consistent with the level effect.

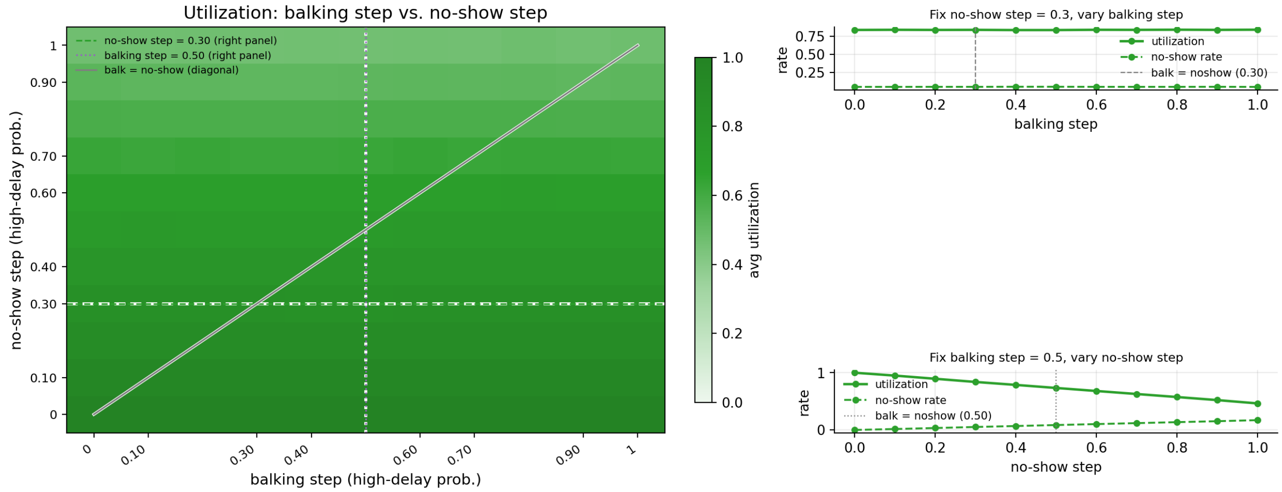

Figure 6: Left: average utilization as a function of balking step (x-axis) and no-show step (y-axis), both classes symmetric. Dashed horizontal and dotted vertical lines mark the slice positions for the right panels; the diagonal marks `balk_step` = `noshow_step`. Top right: utilization and no-show rate as balking step varies with no-show step fixed at 0.3—nearly flat, confirming that the balking step has little direct effect on utilization once no-show step is held constant. Bottom right: the same metrics as no-show step varies with balking step fixed at 0.5—a steep monotone decline driven by wasted booked capacity.

Threshold ordering. Reparametrizing the two thresholds into level and difference reveals a structurally distinct effect from the step analysis. The average threshold level carries a positive coefficient for utilization ($\beta = +0.296$), served rate ($\beta = +0.139$), and offered wait ($\beta = +0.148$): pushing both thresholds out in days reduces the share of accepted patients exposed to behavioral risk zones. The difference (balking threshold minus no-show threshold) carries a significant *negative* coefficient for utilization ($\beta = -0.259$) and served rate ($\beta = -0.158$), and a positive one for offered wait ($\beta = +0.173$).

The mechanism is structural. When balking threshold exceeds no-show threshold (as at baseline: $\theta^{\text{balk}} = 9 > \theta^{\text{no-show}} = 6$), a *toxic zone* opens for $\tau \in (\theta^{\text{no-show}}, \theta^{\text{balk}}]$: patients in this delay range accept offers (no balking risk yet) but carry positive no-show probability. These slots are booked and then wasted at rate ξ^{high}. When the thresholds invert ($\theta^{\text{no-show}} > \theta^{\text{balk}}$), the toxic zone vanishes: patients reach the balking threshold before the no-show threshold, so those who would have been in the high-no-show zone either reject early (balk) or, if they accept, have not yet entered the no-show region. Crucially, the sign of the difference is itself significant for utilization ($\beta = -0.198$, $p = 0.003$), indicating a qualitative regime change at the crossover—not merely a continuous slope. This contrasts with the step dominance effect, where the sign was not significant.

Figure 7 shows the heatmap and projections. The gradient is strong along both axes but asymmetric about the diagonal: above the diagonal (no-show threshold > balking threshold) utilization is higher for a given average threshold level. The top projection (fix no-show threshold at 6 d, vary balking threshold) is nearly flat—once the no-show threshold is held fixed, moving the balking threshold around it has little additional effect. The bottom projection (fix balking threshold at 9 d, vary no-show threshold) shows a steep rise as the no-show threshold increases past the balking threshold, confirming the regime-change interpretation.

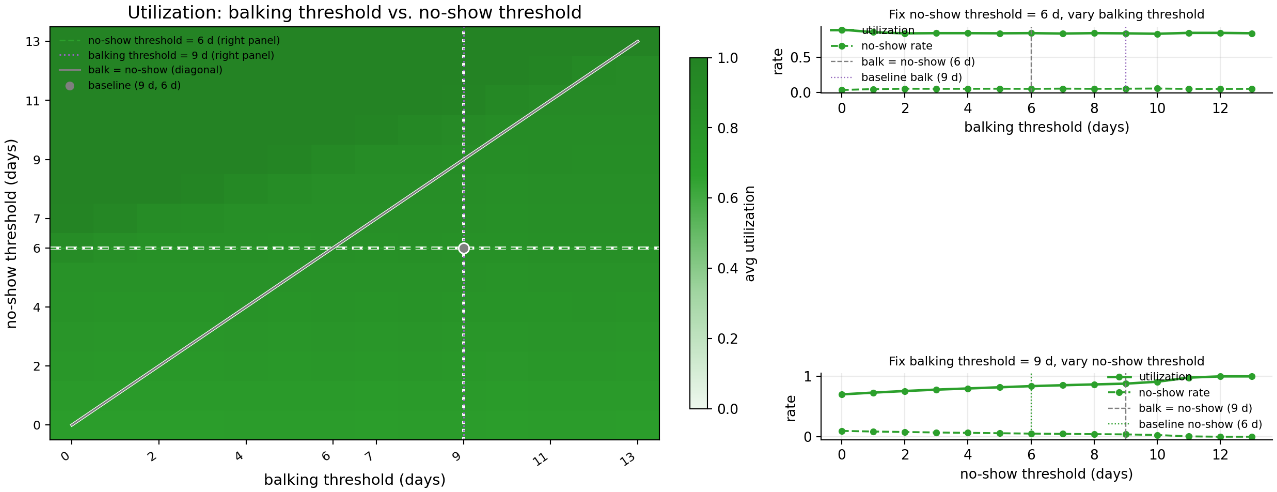

Figure 7: Left: average utilization as a function of balking threshold (x-axis) and no-show threshold (y-axis), both thresholds symmetric across classes. The filled circle marks the baseline ($\theta^{\text{balk}} = 9$ d, $\theta^{\text{no-show}} = 6$ d), which lies below the diagonal in the toxic-zone regime. Above the diagonal the no-show threshold exceeds the balking threshold, eliminating the accept-then-miss zone and raising utilization. Top right: fix no-show threshold at 6 d—nearly flat as balking threshold varies, confirming that the toxic zone width (not the balking threshold level alone) drives the effect. Bottom right: fix balking threshold at 9 d—utilization rises steeply as no-show threshold increases past the balking threshold (at the dashed vertical line), then flattens once the toxic zone is closed.

Class gap drivers. In the randomized screen, the largest class-gap drivers are cancellation probability gap ($|\beta| = 0.452$), balking threshold gap ($|\beta| = 0.429$), and balking step gap ($|\beta| = 0.349$). These findings are plausible: in a pooled FCFS system, class disparities arise because one class accepts more long-delay offers or is more likely to cancel after booking.

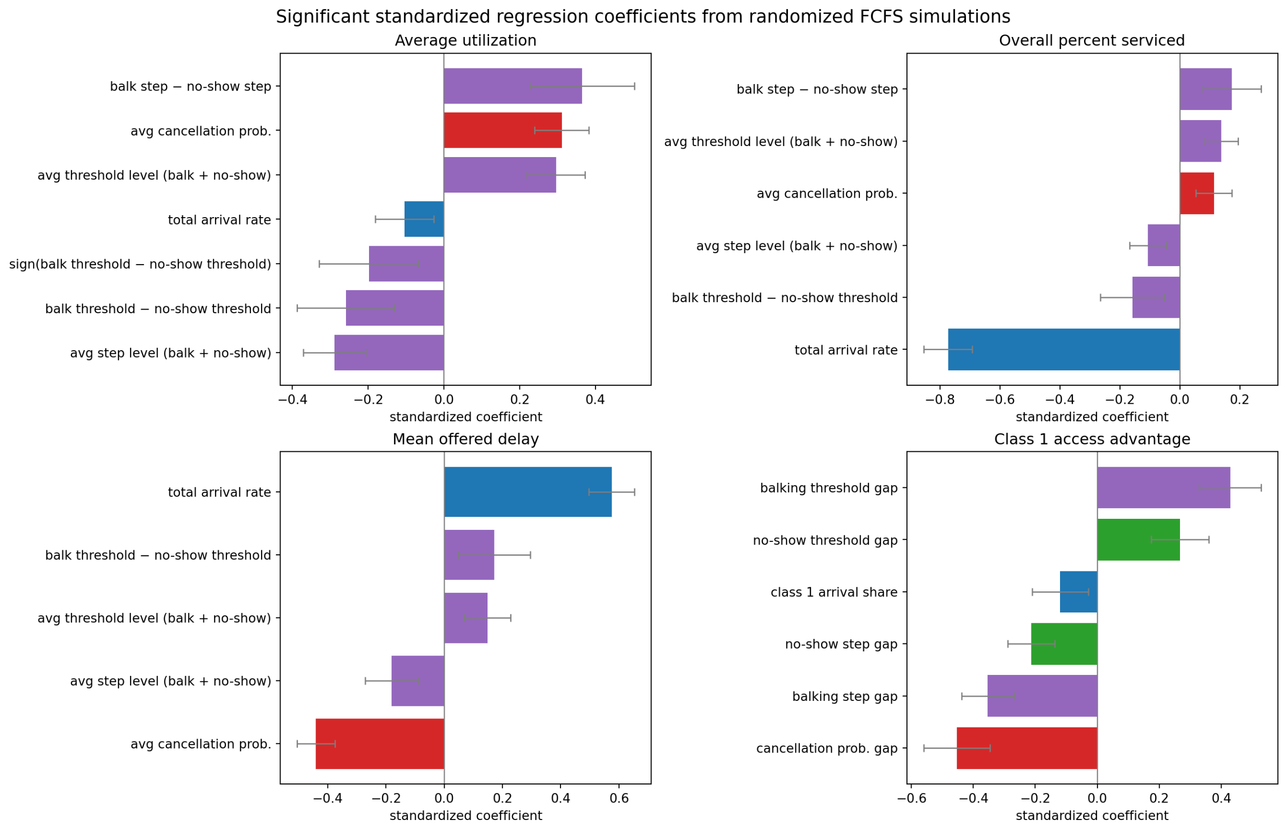

Figure 8: Significant standardized regression coefficients from 240 randomized parameter settings. Features and targets are standardized. The dominance of total arrival rate for served rate and offered wait is consistent with the structural decoupling of utilization and access. Error bars are 95% HC3 robust confidence intervals. Full table in Appendix D.

7 Balking Deep Dive: Selection vs. Access

This section isolates Class 1 balking while Class 2 stays at baseline, to test whether lower accepted wait can accompany lower access.

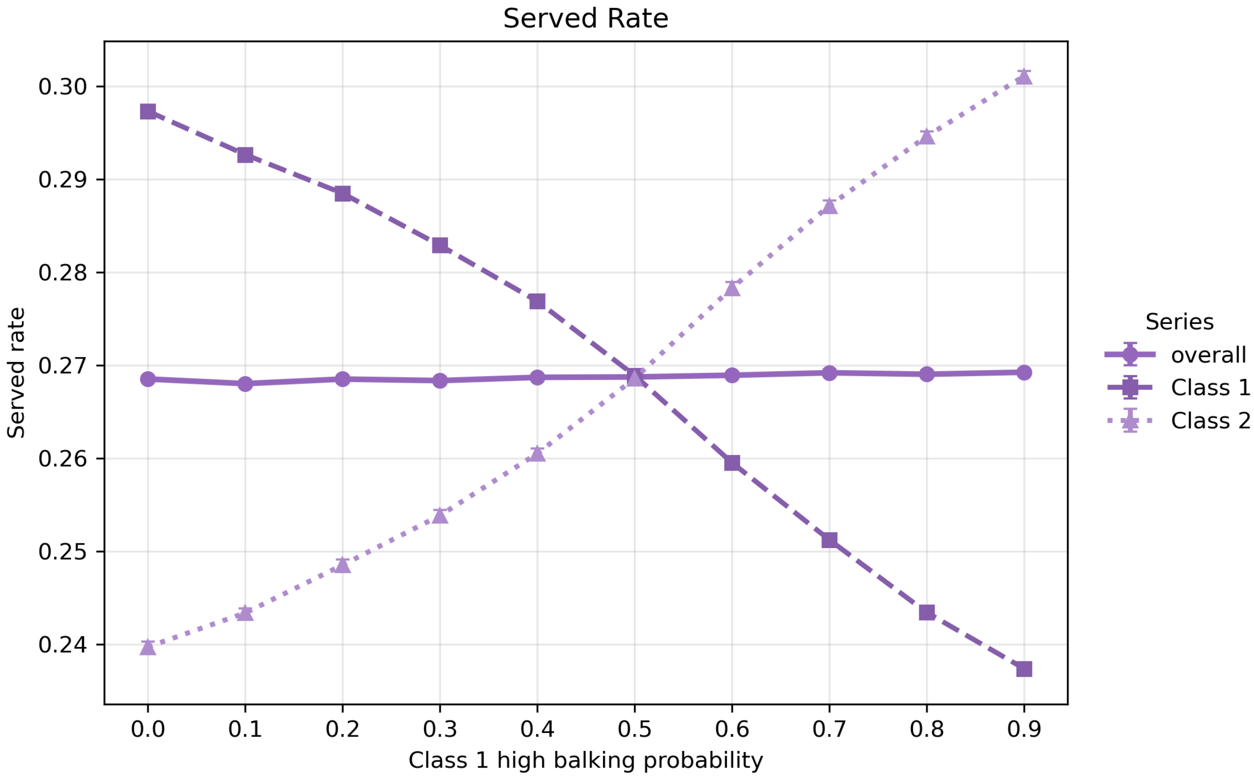

Figure 9: As Class 1 high-balking probability increases from 0 to 1, Class 1 served rate falls while Class 2 served rate rises. The congestion relief that Class 2 gains is mechanically consistent with fewer Class 1 patients booking long-delay slots.

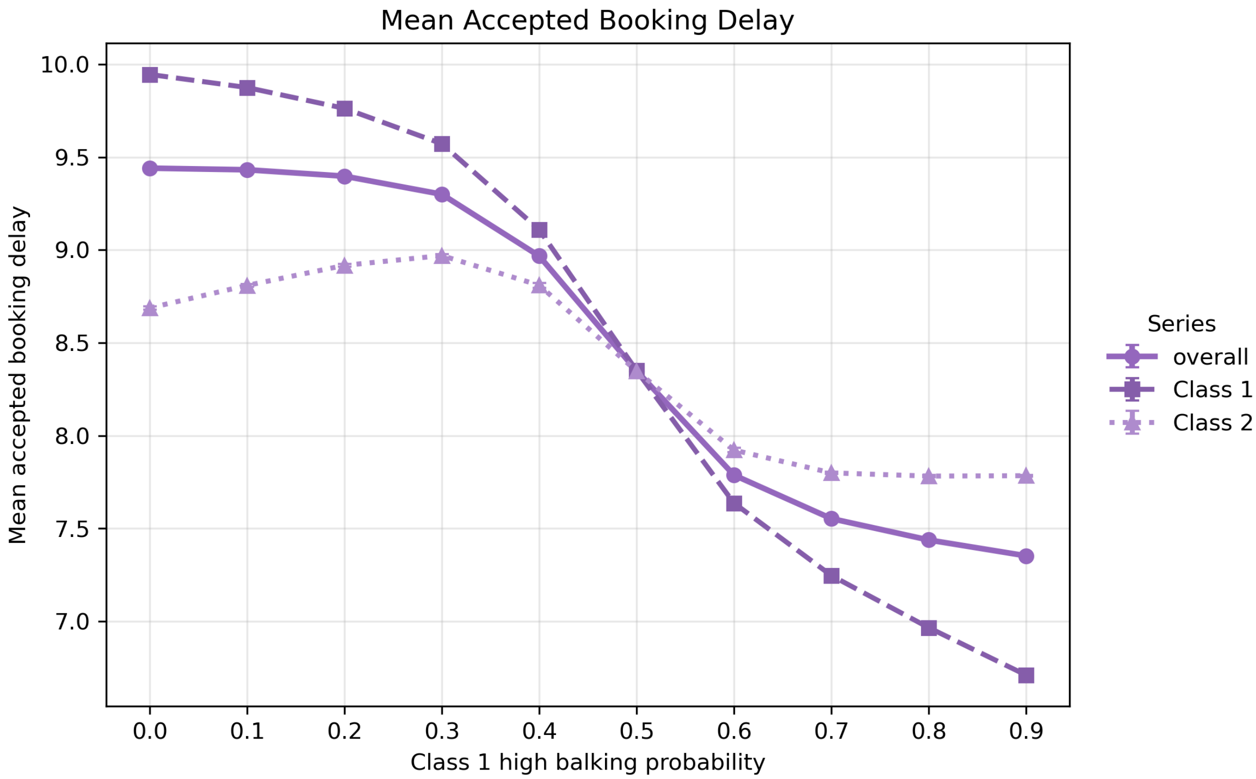

Figure 10: Class 1 accepted wait falls as Class 1 high-balking probability increases. This is a selection artifact: long-delay Class 1 patients leave the accepted pool. Accepted wait should not be read as a delay-improvement signal in this context.

The two figures together demonstrate a pattern the formulation produces cleanly: accepted wait and served rate move in opposite directions when balking changes. Any policy that raises balking step would show apparently improved wait times while actually reducing the share of Class 1 patients who complete service.

8 Class-Gap Interpretation Under Pooled FCFS

Arrivals from both classes are pooled and randomly permuted each day before FCFS assignment. The engine does not distinguish between classes when offering slots. Therefore:

- When class parameters are symmetric, the expected class served-rate gap is zero. Observed deviations are simulation noise.
- Meaningful class gaps arise only from asymmetric behavioral parameters (different balking threshold, balking step, cancellation probability, or no-show behavior) or from different arrival rates. These create systematic differences in how often each class successfully books, cancels, and attends.
- A positive class 1 balking threshold relative to class 2 means class 1 tolerates longer delays before rejecting offers. Under FCFS, a more tolerant class captures more of the later-horizon slots and therefore achieves a higher served rate.

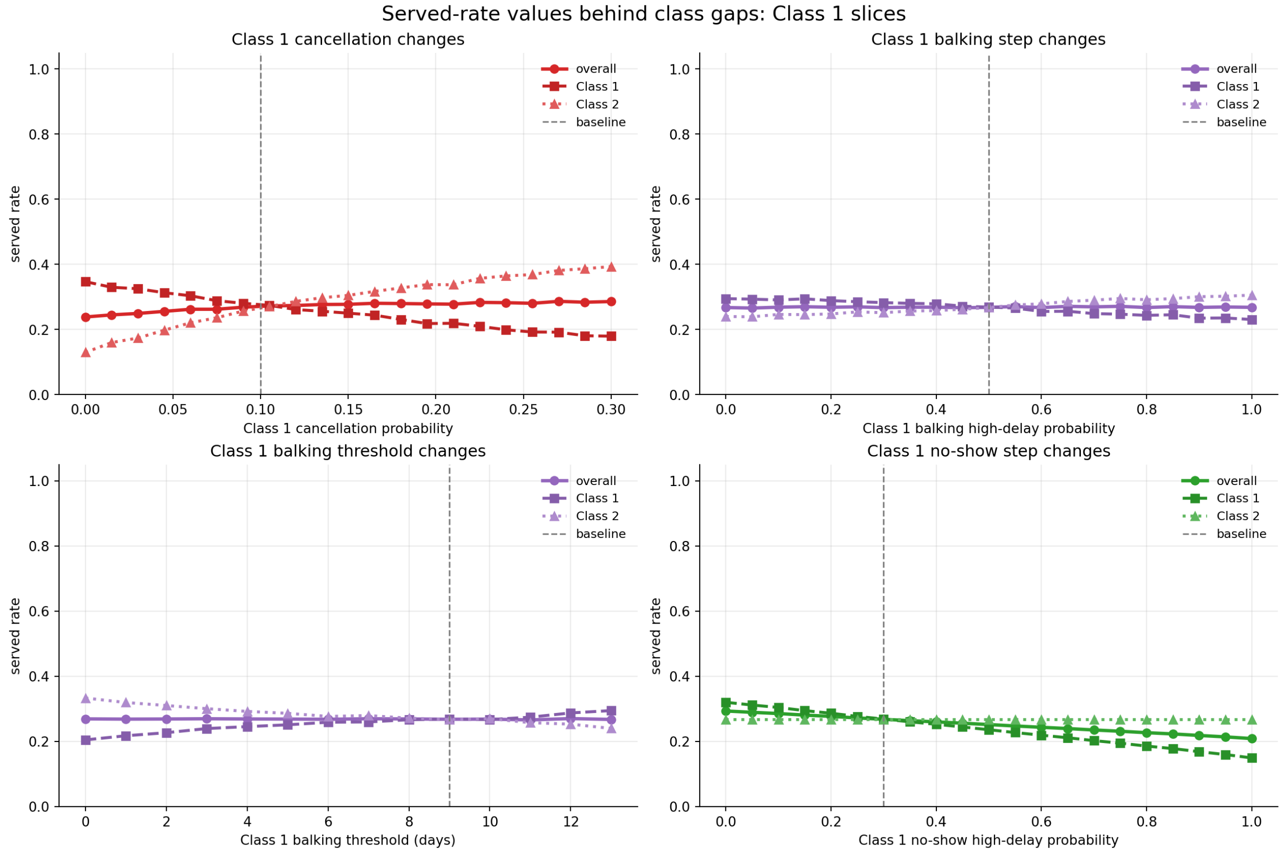

Figure 11: When Class 1 behavioral parameters are varied while Class 2 stays at baseline, a gap opens. Higher Class 1 cancellation, higher balking step, or higher no-show step widen the gap against Class 1. Higher Class 1 balking threshold helps Class 1 by allowing it to accept longer-delay slots. With symmetric parameters, both class lines coincide.

The regression screen confirms this ranking: among class-gap drivers, cancellation probability gap dominates ($|\beta| = 0.452$), followed by balking threshold gap ($|\beta| = 0.429$) and balking step gap ($|\beta| = 0.349$). Class 1 arrival share enters significantly ($\beta = -0.120$): a larger Class 1 share tends to mildly reduce the Class 1 served-rate advantage, consistent with more Class 1 arrivals competing for the same slots under FCFS.

9 What Seems Mechanically Robust vs. What Needs Validation

9.1 Mechanically Robust Under the Current Formulation

- **Utilization and served rate have different denominators.** Their decoupling is a direct consequence of the metric definitions. It is not a fragile numerical result.
- **High utilization can coexist with low served rate.** When demand exceeds supply, served rate is bounded by the supply-to-demand ratio. This bound holds regardless of behavioral parameters.
- **Offered wait is less selective than accepted wait.** Offered delay is recorded before balking, so offered wait includes patients who ultimately leave. Accepted wait excludes them. The direction of the difference is structural.

- **No-offer is a separate access failure not captured by offered wait.** No-offer patients receive no delay information and are excluded from all delay metrics. Above $\lambda \approx 110$, they become a substantial fraction of arrivals.

- **Class labels are exchangeable under symmetric parameters.** FCFS pooling and random permutation mean that class identity has no systematic effect when classes are identical. The near-zero baseline class gap (0.001) is consistent with this.

- **The cancellation–delay feedback is directionally correct.** Longer average booking delay creates more cancellation-check opportunities, which raises cumulative cancellation probability. The high observed cancellation fraction at long delays is mechanically consistent with this.

- **No-offer is zero at baseline; access failures are behavioral.** The booking horizon does not saturate at $\lambda = 100$ because cancellations continuously free future capacity. Balking and cancellation, not horizon saturation, are the primary failure modes.

9.2 Needs Validation or Further Code Checks

1. **Confirm cancellation applies once per residual day.** The documentation says cancellation is checked each day for all future appointments. If the engine instead applies cancellation once at booking, the high cancellation fraction is unexplained and a code audit is needed. Check `simulation/engine.py` cancellation loop.

2. **Clarify no-offer propagation within a day.** The documentation states that when the horizon is full for one patient, all remaining arrivals that day are counted as `no_offer`. This means a single saturation event affects the full day remainder, which could create day-order artifacts. Verify whether the no-offer share is sensitive to the random permutation seed.

3. **Assess whether no-show slots should remain unbooked.** No-show slots are not rebooked in the current formulation. In many real clinics, same-day no-shows are filled with walk-ins or advance overbooking. If this assumption is changed, both utilization and served rate would shift; the magnitude depends on the no-show fraction and same-day availability.

4. **Assess whether same-day cancellations should be excluded.** Currently, no same-day cancellations are allowed. This keeps service-day slots deterministic once the service day is reached. Relaxing this would add a slot-recapture mechanism that could affect both utilization and served rate.

5. **Confirm stability of class-gap estimates.** The class-gap metric has the lowest out-of-sample R^2 (0.708) in the regression screen and is near zero (0.001) at symmetric baseline. With only 2 seeds per setting in the regression screen, sampling variance in the gap could be material. At least 10–20 seeds per grid point are advisable before drawing strong conclusions about class equity.

6. **Evaluate realism of baseline demand and capacity levels.** 100 arrivals per day against 32 slots per day is an intentional stress-test, not a calibrated clinic. Before drawing any operational conclusions, behavioral parameters (balking thresholds, no-show probabilities, cancellation rates) should be estimated from clinic data and the demand-to-capacity ratio brought into a plausible range.

10 Summary for Discussion

1. Under the current formulation, utilization and served rate are structurally decoupled. At baseline, this produces a 3:1 discrepancy (0.839 vs. 0.269). Using utilization as a proxy for access would be misleading at this demand level.

2. The dominant loss channels at baseline are balking (35.6%) and cancellation (32.3%). No-offer is zero. Above $\lambda \approx 110$, horizon saturation becomes an additional channel.

3. The high cancellation fraction is consistent with the long average booking delay and the daily-cancellation rule. Whether this is realistic depends on how real patients behave when booked many days out.

4. Offered wait is preferable to accepted wait as a delay measure, because it includes patients who rejected the offer. But it still excludes no-offer patients, who are the most severely access-constrained.

5. Class disparities in the simulation arise from asymmetric behavioral parameters, not from FCFS favoritism. Symmetric classes produce near-zero gaps.

6. The regression screen and one-at-a-time sensitivity runs are internally consistent: demand pressure drives served rate and offered wait; no-show and cancellation behavior drive utilization; behavioral parameter gaps drive class equity.

7. None of these findings should be interpreted as causal estimates about real appointment systems. They describe how this particular model formulation responds to parameter variation.

Questions for discussion:

1. Is the baseline demand/capacity ratio intended as a stress scenario, or should it be recalibrated toward a more plausible clinic setting?

2. Should cancellation be modeled as a daily hazard (once per residual day) or a one-time event at booking? The cumulative cancellation fraction depends critically on this.

3. Should no-offer patients be reported as a separate access failure in all summary tables, not only when the no-offer rate is non-negligible?

4. Should no-show probability depend on the original offered delay τ (current formulation) or on the remaining wait at service time?

5. Should same-day cancellations be allowed? What would the rebooking mechanism look like?

6. Should alternative scheduling rules (e.g., priority-based, overbooking) be compared against FCFS within this formulation?

A Exact Metric Definitions and Notation

Let i be patient class, D measured service days, S slots per day, A_i arrivals, B_i accepted bookings, K_i balked patients, C_i cancellations, H_i no-shows, V_i completed visits, and $O_i = B_i + K_i$ offered patients.

$$\begin{aligned}
\texttt{percent_serviced}_i &= V_i/A_i,\\
\texttt{overall_percent_serviced} &= \textstyle\sum_i V_i \;/\; \sum_i A_i,\\
\texttt{mean_offered_booking_delay} &= \textstyle\sum_i \tau_i^{\text{off}} \;/\; \sum_i O_i,\\
\texttt{mean_accepted_booking_delay} &= \textstyle\sum_i \tau_i^{\text{acc}} \;/\; \sum_i B_i,\\
\texttt{average_utilization} &= \tfrac{1}{D}\textstyle\sum_d V_d/S,\\
\texttt{balking_rate} &= \textstyle\sum_i K_i \;/\; \sum_i O_i,\\
\Delta_{\text{access}} &= \texttt{percent_serviced}_1 - \texttt{percent_serviced}_2.
\end{aligned}$$

Note on terminology. In the main text, `percent_serviced` (the simulation's internal variable name) is reported as *overall served rate*. `mean_offered_booking_delay` is reported as *mean offered wait*; `mean_accepted_booking_delay` as *mean accepted wait*.

Behavioral step functions.

$$b_i(\tau) = \begin{cases} b_i^{\text{low}} & \tau < \theta_i^{\text{balk}},\\ b_i^{\text{high}} & \tau \ge \theta_i^{\text{balk}}.\end{cases} \qquad \xi_i(\tau) = \begin{cases} \xi_i^{\text{low}} & \tau < \theta_i^{\text{no-show}},\\ \xi_i^{\text{high}} & \tau \ge \theta_i^{\text{no-show}}.\end{cases}$$

Balking step $= b_i^{\text{high}} - b_i^{\text{low}}$; no-show step $= \xi_i^{\text{high}} - \xi_i^{\text{low}}$.

Measurement semantics. Class metrics cover patients arriving in the measurement window. If `cooldown_days` $<$ `horizon_days` $- 1$, some late bookings are unresolved at simulation end and counted as `unresolved_booked` (0.06% of arrivals at baseline). `total_value` is service-day based and is not cohort-aligned with class metrics.

B Simulation Design and Sensitivity Grid

All sensitivity grids are centered on `configs/baseline.yaml`. One-axis slices vary Class 1 parameters while Class 2 stays at baseline. Two-class heatmaps vary both classes independently.

Parameter	Baseline	Range	Grid size
Arrivals (per class), 1D slices	50	25–75	continuous
Total arrivals, capacity stress	100	30–170	continuous
Balking step	0.50	0.00–1.00	21
Balking threshold (days)	9	0–13	14
No-show step	0.30	0.00–1.00	21
No-show threshold (days)	6	0–13	14
Cancellation probability	0.10	0.00–0.30	21

Table 4: Sensitivity parameter ranges. Each grid point uses seed 2027; multi-seed comparisons average seeds 2027, 2028, 2029.

Regression screen. 240 randomized parameter settings (seed 9137); two seeds (4101, 4102) averaged per setting; 12 features (total arrival rate, Class 1 share, average level and Class 1−Class 2 gap for each behavioral parameter); OLS with HC3 robust standard errors; only $p < 0.05$ coefficients shown.

Balking deep dives. Class 1 high-balking-probability sweep over $\{0.0, 0.1, \ldots, 0.9\}$ (100 seeds per point; Class 2 fixed at baseline). Class 1 threshold sweep over $\{3, 4, \ldots, 13\}$ (100 seeds per point; Class 2 fixed at baseline).

C Supplementary Sensitivity Plots

C.1 Driver Plots

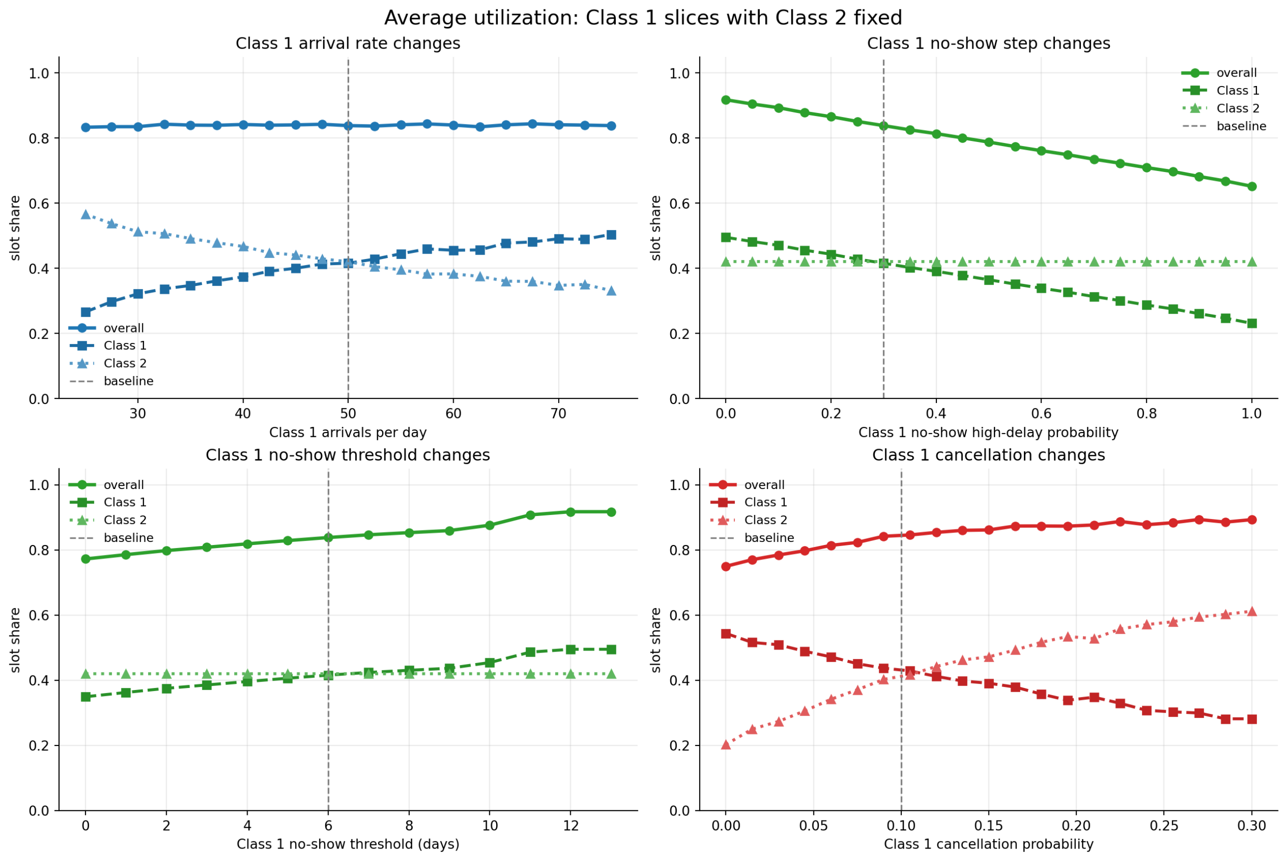

Figure 12: Utilization across Class 1 arrival and behavioral parameter ranges. Utilization stays near 0.839 across most of the demand range, while class-level served rates diverge.

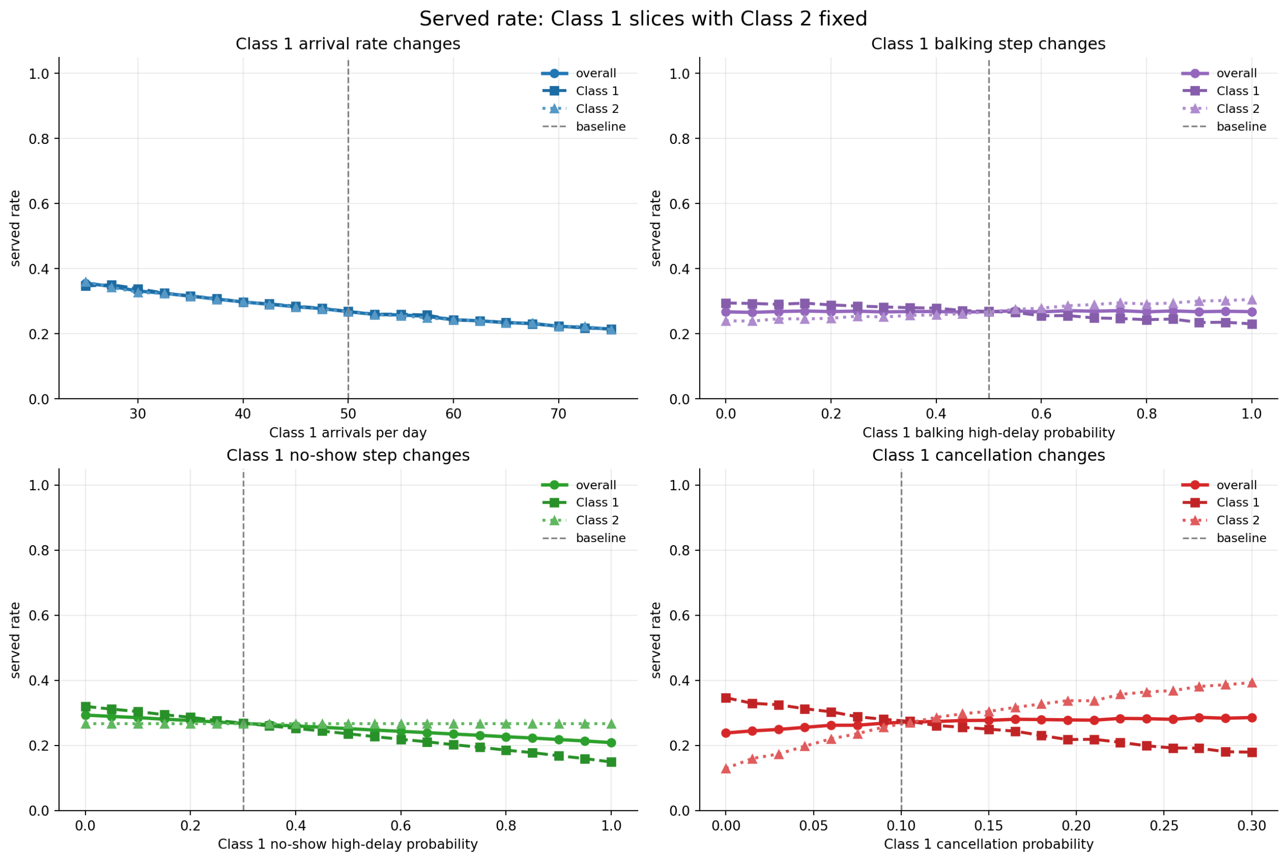

Figure 13: Served rate falls as arrival rates rise. The Class 1 and Class 2 lines separate when Class 1 parameters diverge from symmetric baseline.

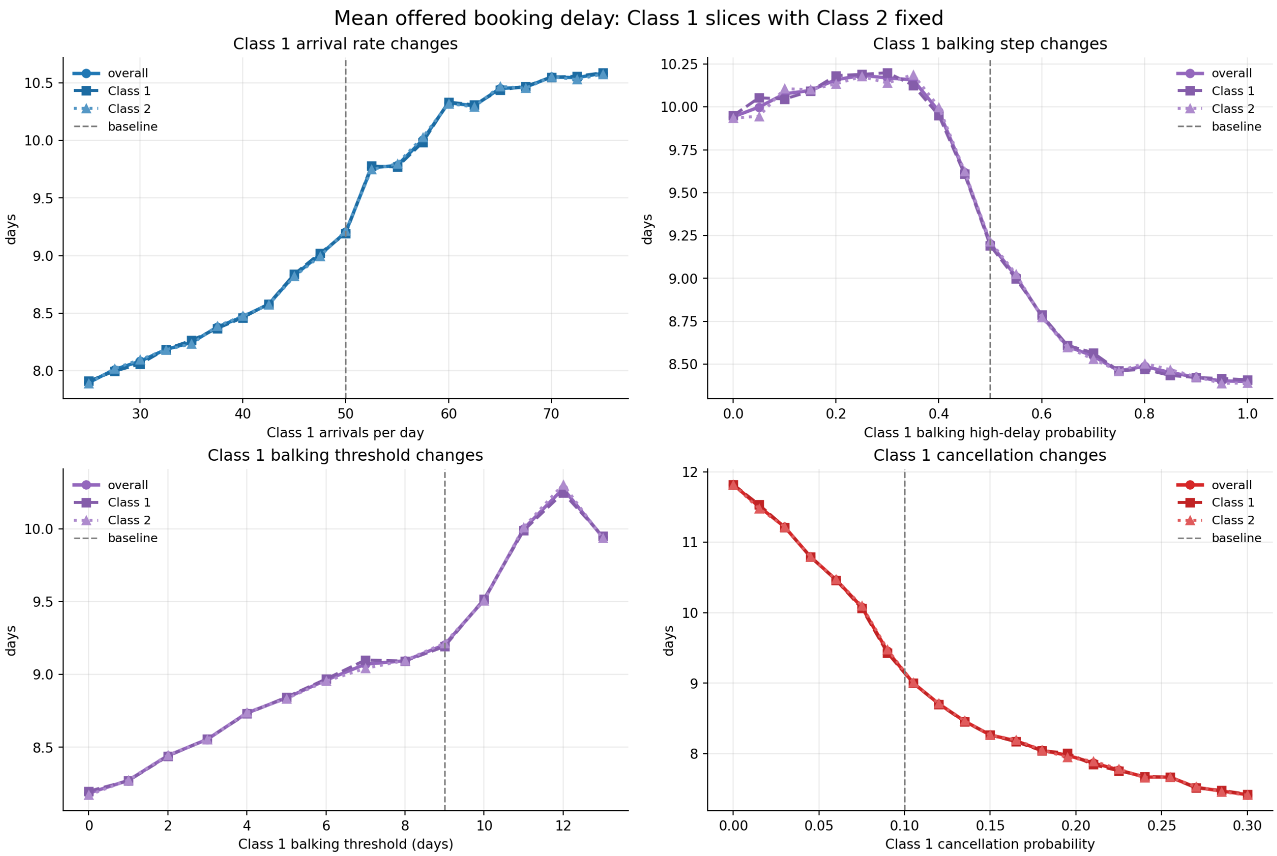

Figure 14: Offered and accepted wait across Class 1 parameter ranges. Accepted wait can fall while offered wait stays elevated, illustrating selection bias in the accepted-wait metric.

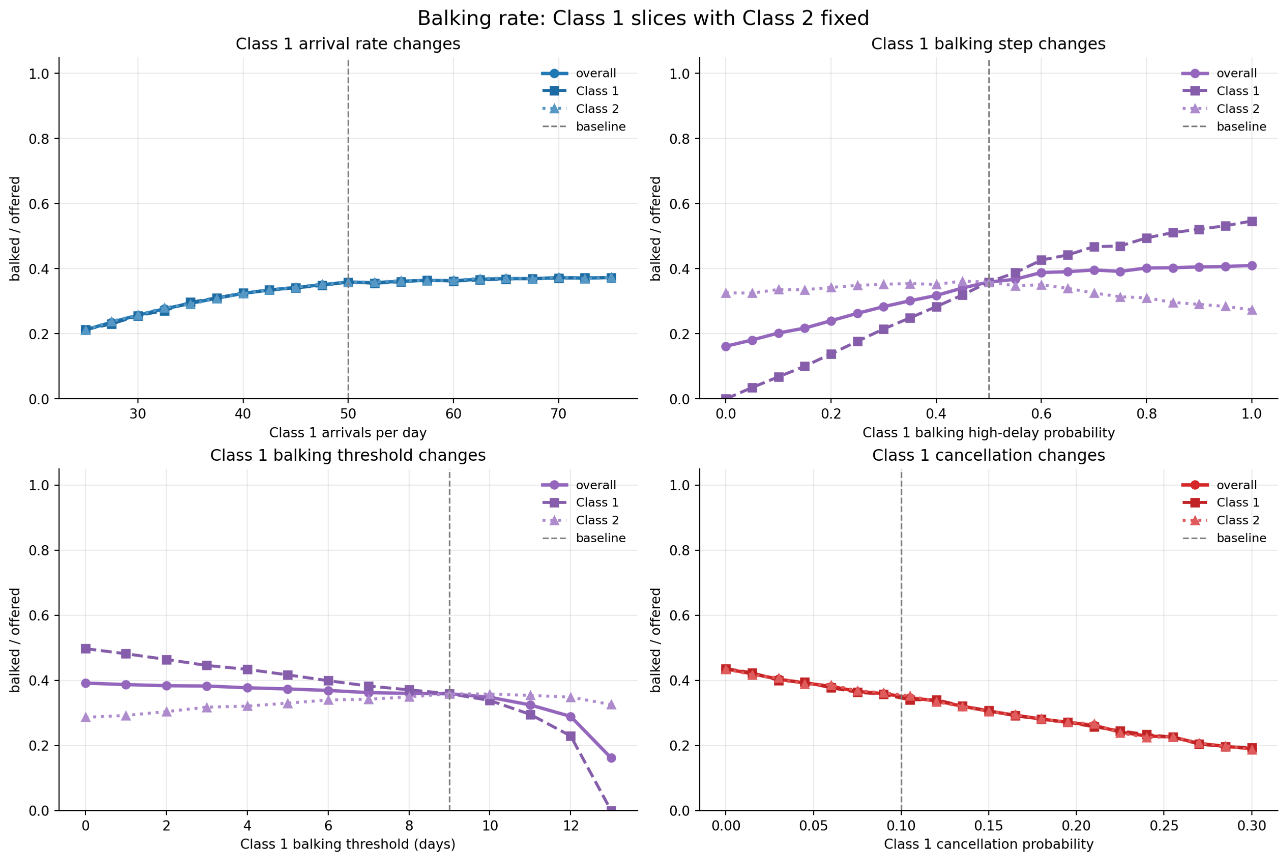

Figure 15: Balking rate is controlled primarily by balking step and threshold. Rising balking rate indicates congestion but not improved service.

C.2 No-Show and Cancellation Heatmaps

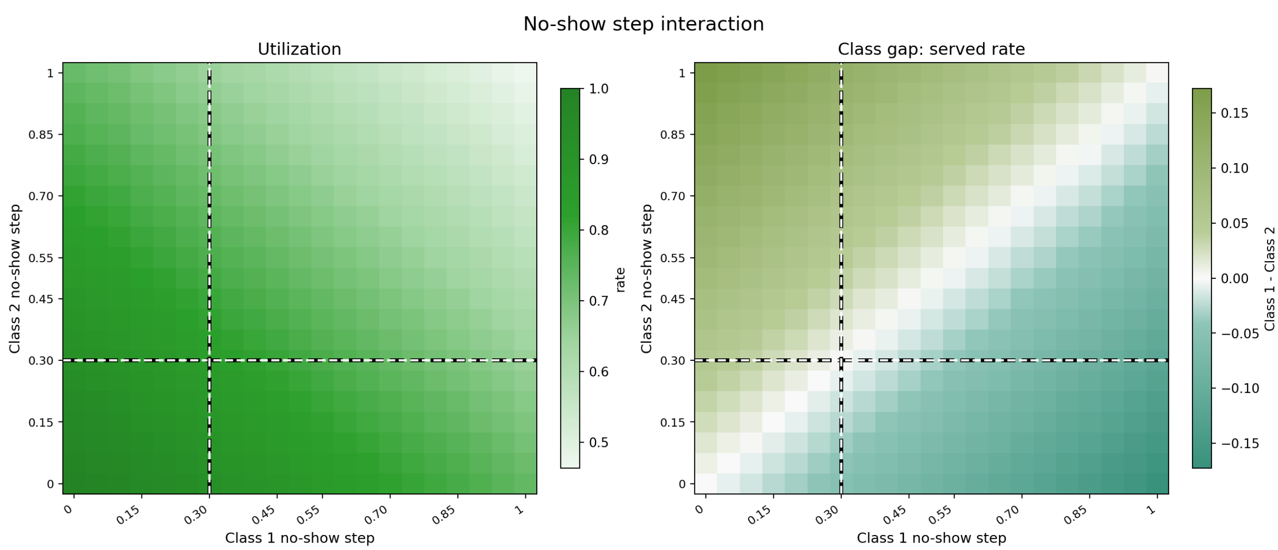

Figure 16: No-show step heatmap. Utilization falls when both classes carry high no-show risk. Class gap moves against the higher no-show class.

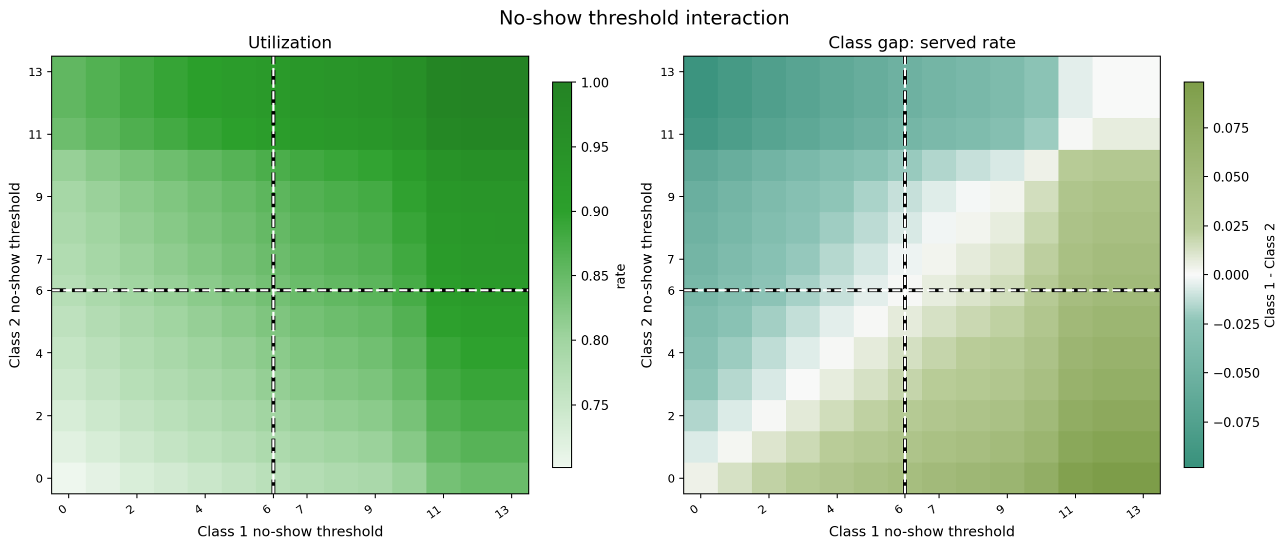

Figure 17: No-show threshold heatmap. Higher threshold delays high no-show onset, improving utilization and access for the affected class.

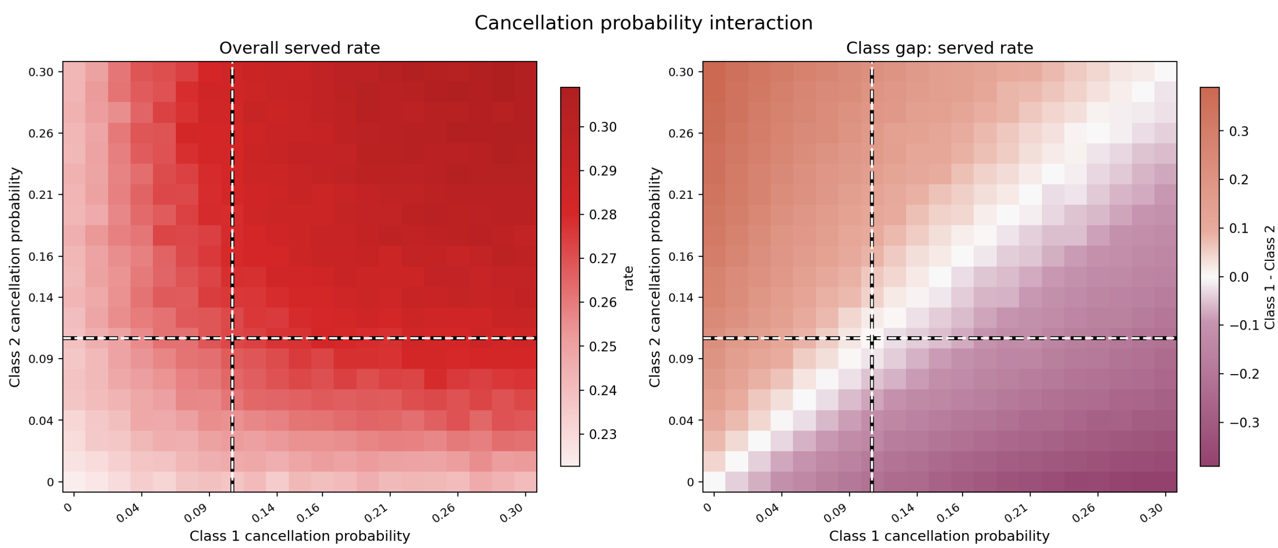

Figure 18: Cancellation heatmap. Higher cancellation for one class reduces that class's access; aggregate effects are more complex (see Section 6).

C.3 Balking Heatmaps

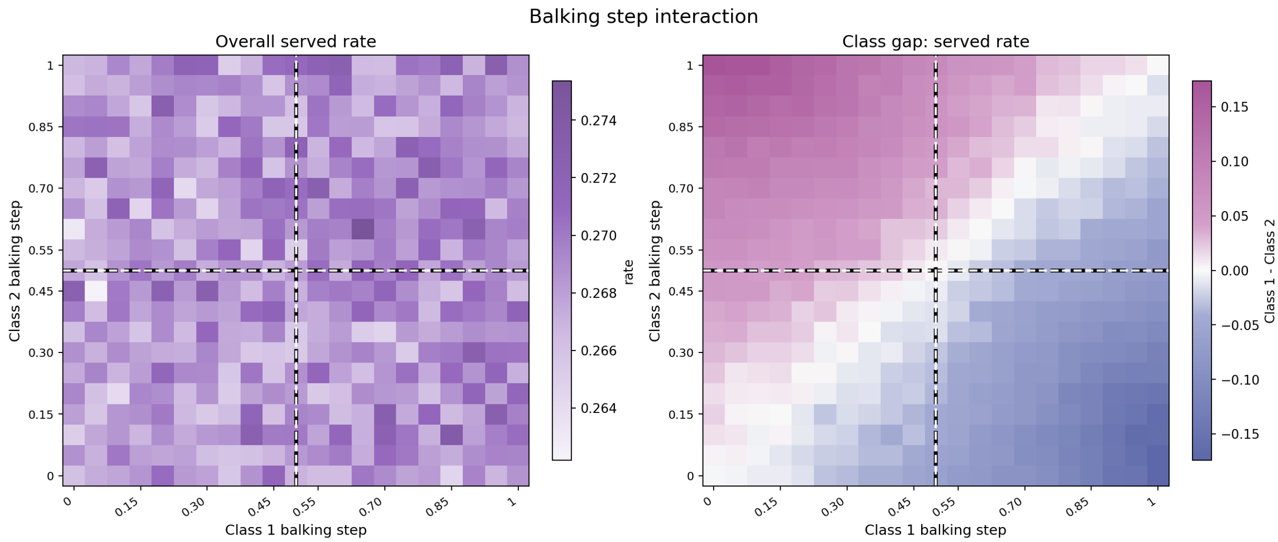

Figure 19: Balking step heatmap. Served rate falls and class gap widens as one class's balking step rises above the other.

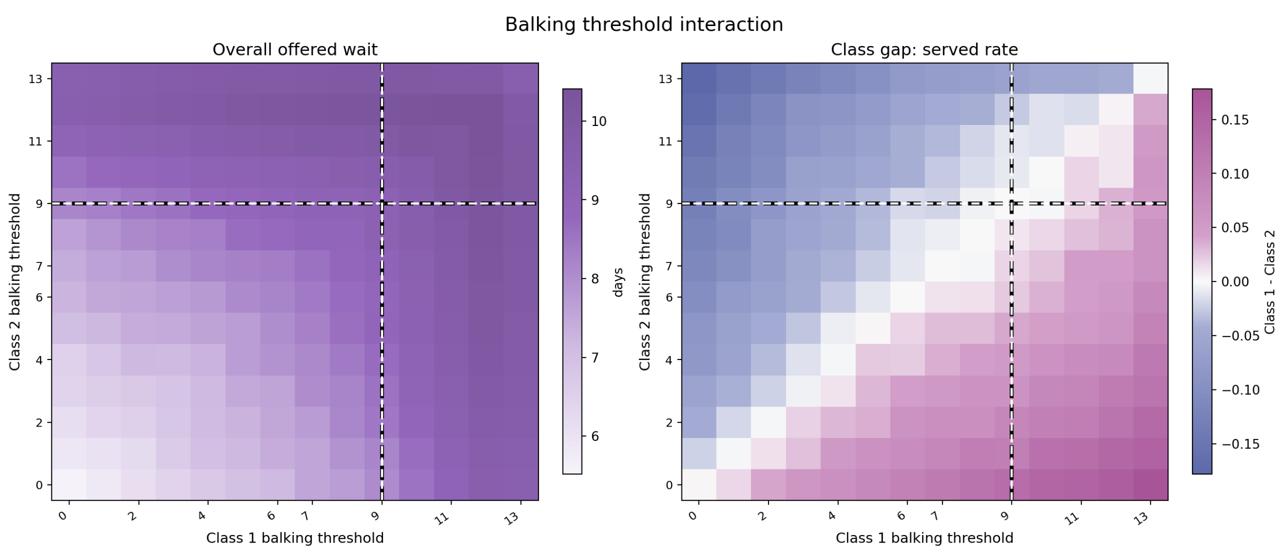

Figure 20: Balking threshold heatmap. The more tolerant class (higher threshold) captures more served slots under FCFS.

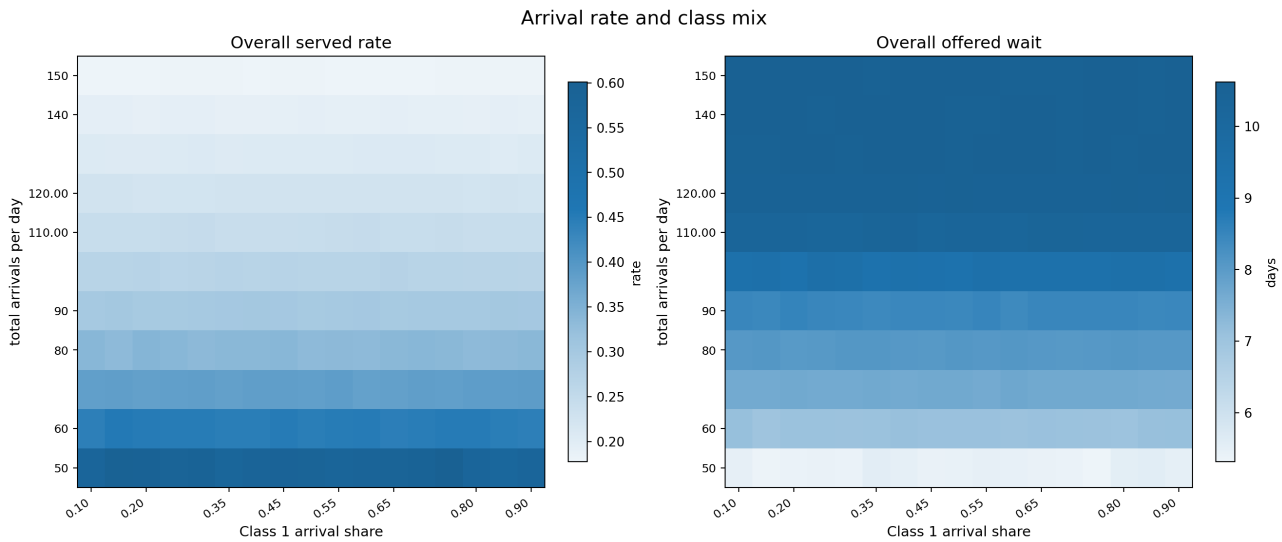

Figure 21: Arrival mix heatmap. Left: overall served rate. Right: mean offered wait.

C.4 Capacity Stress

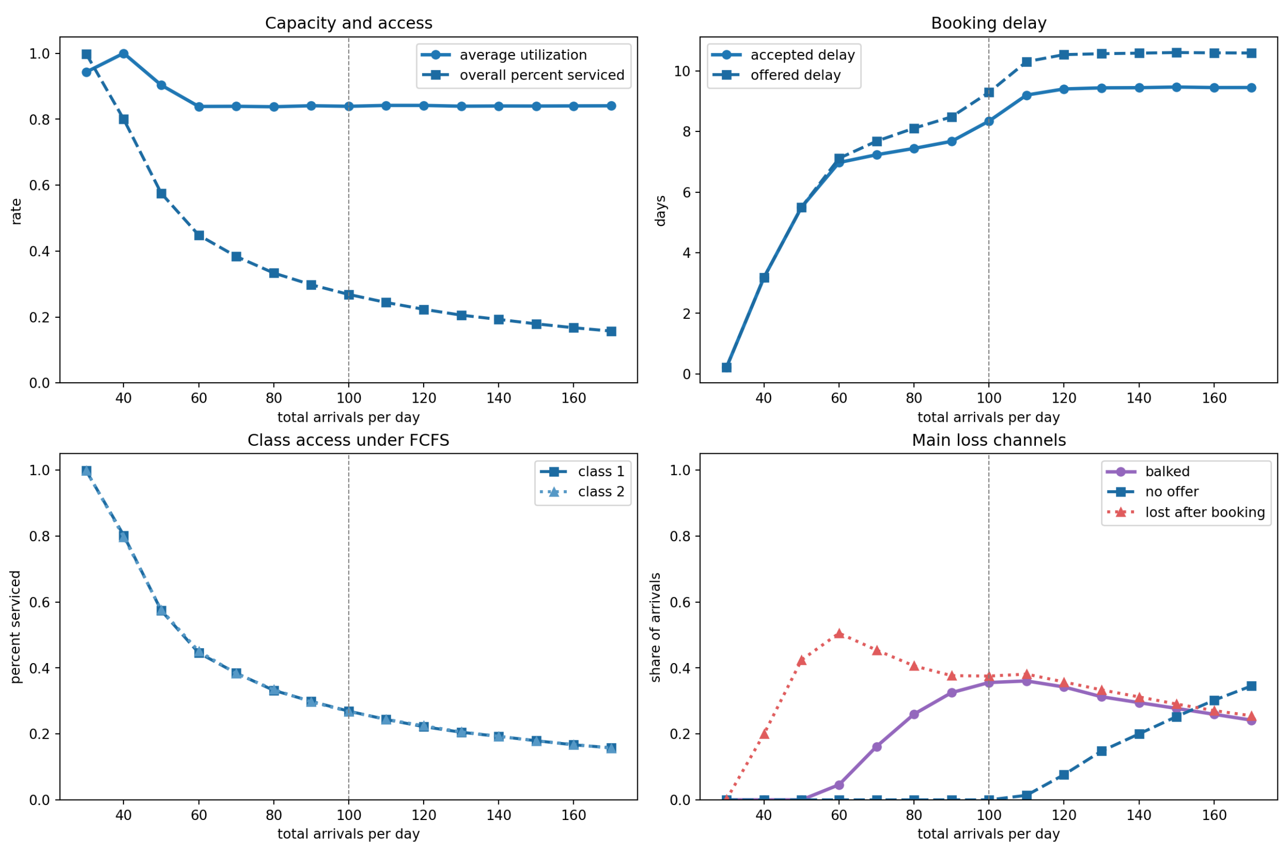

Figure 22: Capacity stress diagnostic curves. Served rate and offered wait as demand rises from $\lambda = 30$ to 170; utilization remains nearly flat across the full range.

C.5 Balking Deep Dive: Additional Plots

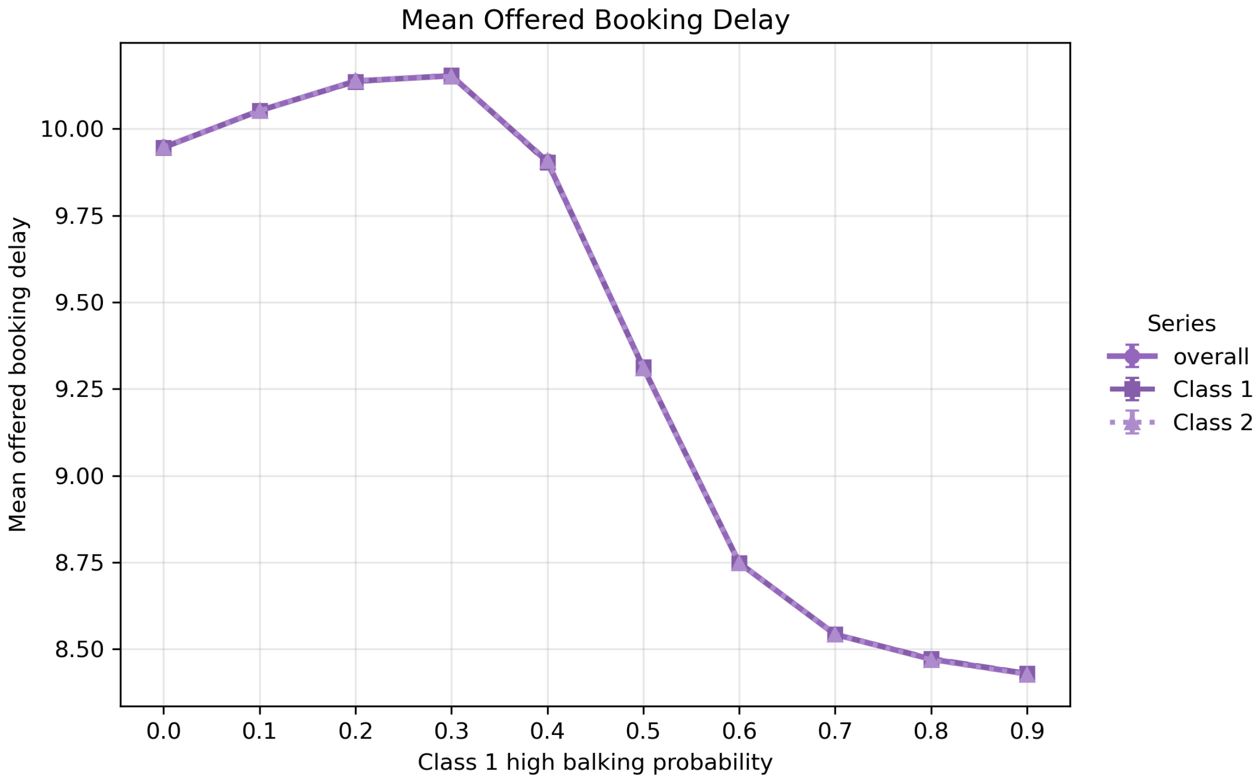

Figure 23: Offered wait by class as Class 1 high-balking probability rises. When enough Class 1 patients reject long-delay offers, overall horizon congestion falls and both classes receive shorter offered delays.

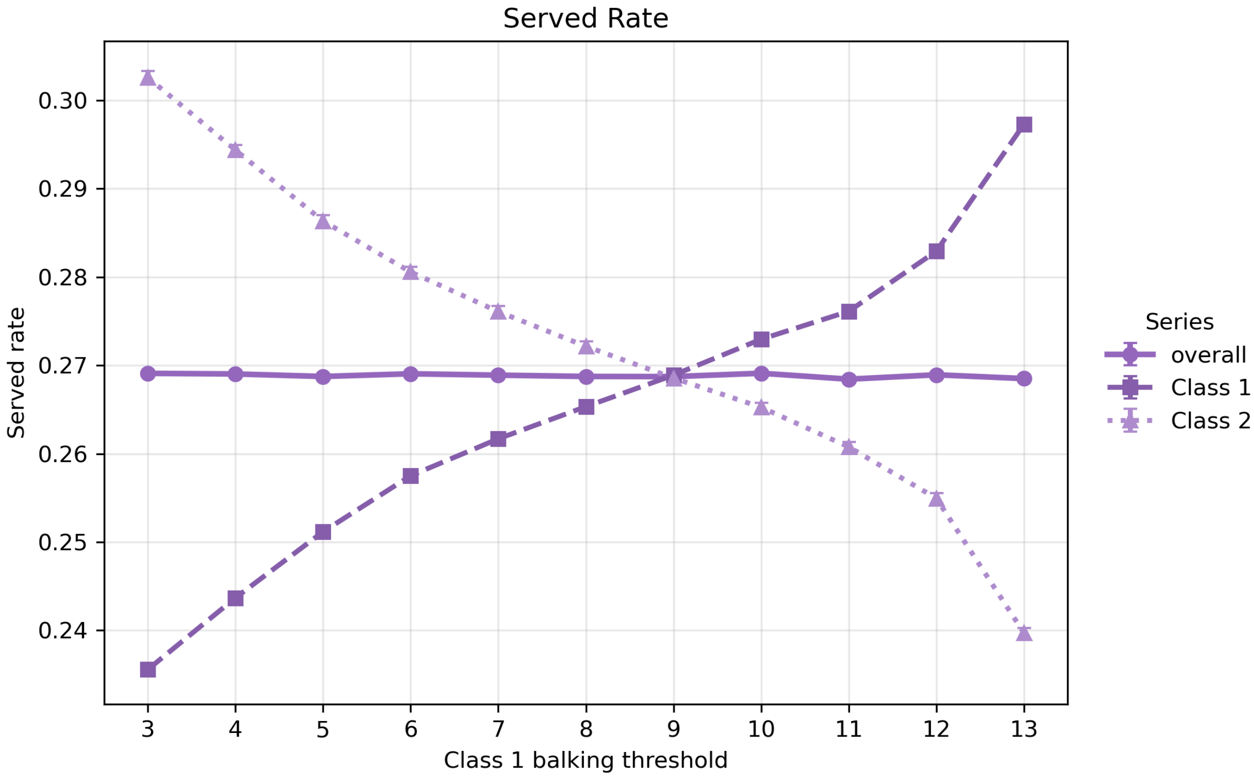

Figure 24: Class 1 balking threshold sweep: higher threshold increases Class 1 access by tolerating longer delays. The gain comes at mild cost to Class 2 when Class 1 tolerance is very high.

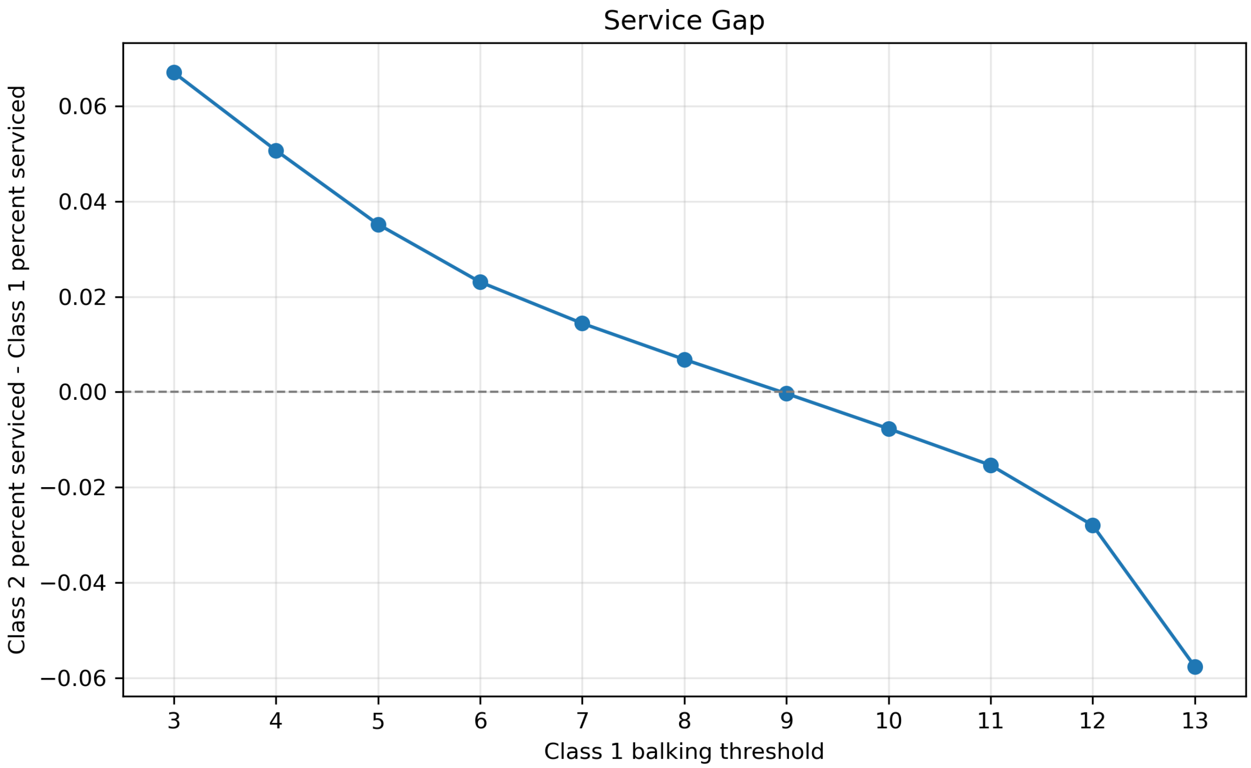

Figure 25: Class 1 minus Class 2 served-rate gap as Class 1 threshold varies.

D Regression Screen

Target	Feature	Std. coef.	95% HC3 CI	HC3 p
Utilization	Avg threshold level (balk + no-show)	$+0.296$	$[0.219,\ 0.373]$	< 0.001
Utilization	Balk step − no-show step	$+0.365$	$[0.227,\ 0.504]$	< 0.001
Utilization	Avg cancellation prob.	$+0.311$	$[0.240,\ 0.384]$	< 0.001
Utilization	Avg step level (balk + no-show)	-0.289	$[-0.371,\ -0.207]$	< 0.001
Utilization	Balk threshold − no-show threshold	-0.259	$[-0.387,\ -0.131]$	< 0.001
Utilization	Sign(balk threshold − no-show threshold)	-0.198	$[-0.330,\ -0.067]$	0.003
Utilization	Total arrival rate	-0.104	$[-0.182,\ -0.027]$	0.008
Overall served rate	Total arrival rate	-0.773	$[-0.854,\ -0.692]$	< 0.001
Overall served rate	Avg threshold level (balk + no-show)	$+0.139$	$[0.084,\ 0.195]$	< 0.001
Overall served rate	Balk step − no-show step	$+0.174$	$[0.077,\ 0.271]$	< 0.001
Overall served rate	Avg step level (balk + no-show)	-0.107	$[-0.168,\ -0.045]$	0.001
Overall served rate	Balk threshold − no-show threshold	-0.158	$[-0.264,\ -0.053]$	0.003
Overall served rate	Avg cancellation prob.	$+0.114$	$[0.054,\ 0.174]$	< 0.001
Mean offered wait	Total arrival rate	$+0.575$	$[0.496,\ 0.653]$	< 0.001
Mean offered wait	Avg cancellation prob.	-0.440	$[-0.504,\ -0.375]$	< 0.001
Mean offered wait	Balk threshold − no-show threshold	$+0.173$	$[0.049,\ 0.296]$	0.006
Mean offered wait	Avg threshold level (balk + no-show)	$+0.148$	$[0.069,\ 0.228]$	< 0.001
Mean offered wait	Avg step level (balk + no-show)	-0.181	$[-0.271,\ -0.091]$	< 0.001
Class gap	Cancellation prob. gap	-0.452	$[-0.558,\ -0.346]$	< 0.001
Class gap	Balking threshold gap	$+0.429$	$[0.331,\ 0.528]$	< 0.001
Class gap	Balking step gap	-0.352	$[-0.436,\ -0.268]$	< 0.001
Class gap	No-show threshold gap	$+0.266$	$[0.175,\ 0.357]$	< 0.001
Class gap	No-show step gap	-0.213	$[-0.289,\ -0.136]$	< 0.001
Class gap	Class 1 arrival share	-0.119	$[-0.209,\ -0.029]$	0.009

Table 5: Significant standardized regression coefficients ($p < 0.05$, HC3 robust SEs). Gap = Class 1 minus Class 2 value. 240 randomized settings, 2 seeds averaged per setting. Only features passing $p < 0.05$ are shown.

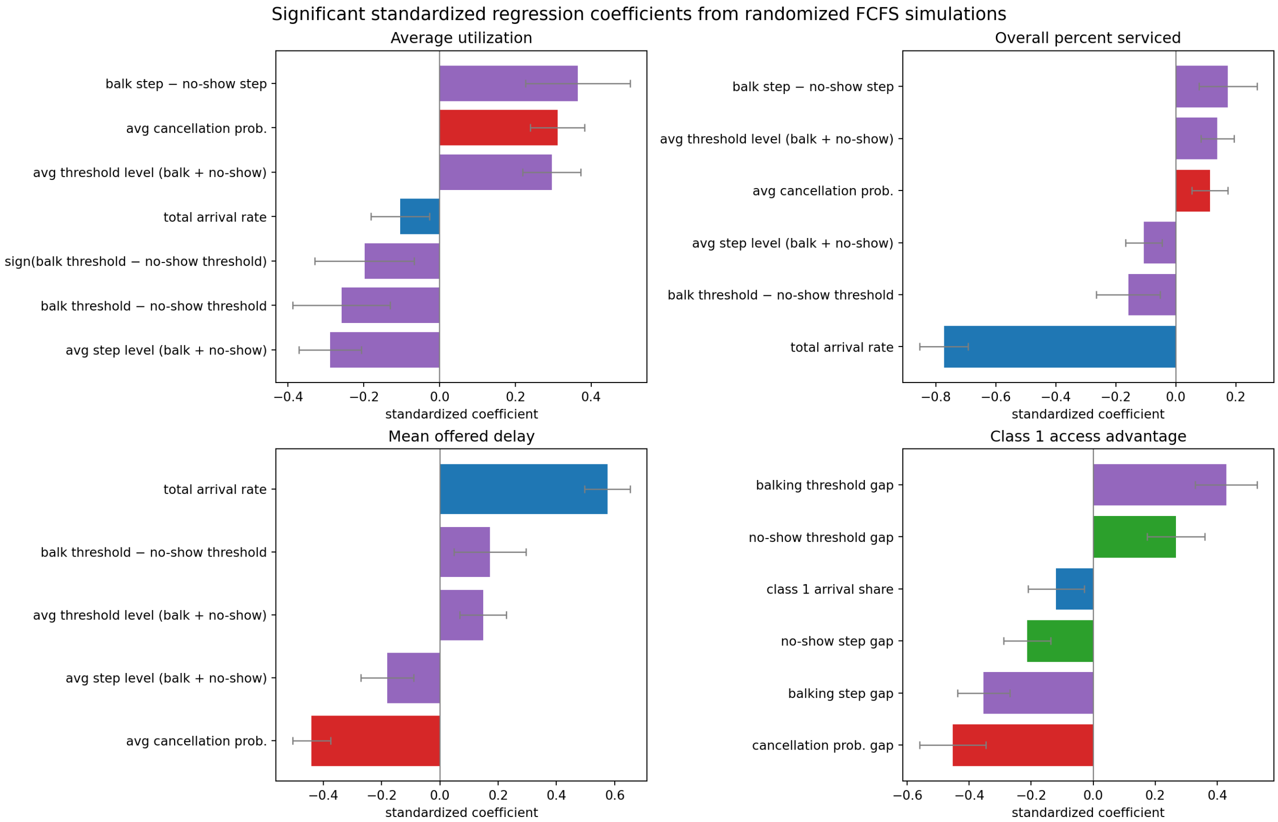

Figure 26: Significant standardized regression coefficients with 95% HC3 confidence intervals. Total arrival rate dominates served rate and offered wait; no-show behavior dominates utilization; behavioral parameter gaps drive the class served-rate gap.